TABLE S1

Kinetic Reactions for HD 189733b and HD 209458b

	Reaction	Rate constant	Reference
R138	$2\,\mathrm{H} \xrightarrow{\mathrm{M}} \mathrm{H_2}$	$k_0 = 2.7 \times 10^{-31}\,T^{-0.6}$	Baulch *et al.* (1994)
R140	$\mathrm{H + CH \longrightarrow C + H_2}$	$1.3 \times 10^{-10}\,e^{(-80/T)}$	Harding *et al.* (1993)
R142	$\mathrm{H + {}^1CH_2 \longrightarrow CH + H_2}$	3.0×10^{-10}	Estimate
R144	$\mathrm{H + {}^3CH_2} \xrightarrow{\mathrm{M}} \mathrm{CH_3}$	$k_0 = 9.0 \times 10^{-32}\,e^{(550/T)}$	Est., $0.1 \times k_0$ for CH + $\mathrm{H_2}$
		$k_\infty = 8.55 \times 10^{-12}\,T^{0.15}$	Est., $0.1 \times k_\infty$ for CH + $\mathrm{H_2}$
R146	$\mathrm{H + CH_3} \xrightarrow{\mathrm{M}} \mathrm{CH_4}$	$k_0 = 1.5 \times 10^{-24}\,T^{-1.8},\ T > 277.5$ K	based on Brouard *et al.* (1989),
		$k_0 = 6.0 \times 10^{-29},\ T \leq 277.5$ K	Seakins *et al.* (1997),
		$k_\infty = 1.92 \times 10^{-8}\,T^{-0.5}\,e^{(-400/T)}$	and Baulch *et al.* (1994)
		$k_\infty = 4.82 \times 10^{-11},\ T \leq 110$ K	
		$F_c = 0.3\ +\ 0.58e^{(-T/800)}$	
R148	$\mathrm{H + CH_4 \longrightarrow CH_3 + H_2}$	$2.2 \times 10^{-20}\,T^3\,e^{(-4045/T)}$	Baulch *et al.* (1992)
R150	$\mathrm{H + C_2H} \xrightarrow{\mathrm{M}} \mathrm{C_2H_2}$	$k_0 = 1.26 \times 10^{-18}\,T^{-3.1}\,e^{(-721/T)}$	Tsang and Hampson (1986)
		$k_\infty = 3.0 \times 10^{-10}$	Tsang and Hampson (1986)
R152	$\mathrm{H + C_2H_2} \xrightarrow{\mathrm{M}} \mathrm{C_2H_3}$	$k_0 = 3.77 \times 10^{-8}\,T^{-7.27}\,e^{(-3633/T)}$	for $T > 400$ K, lit. review
		$k_0 = 3.3 \times 10^{-30}\,e^{(-740/T)},\ T \leq 400$ K	Baulch *et al.* (1994)
		$k_\infty = 7.5 \times 10^{-15}\,T^{1.5}\,e^{(-1300/T)}$	for $T > 400$ K, lit. review
		$k_\infty = 1.4 \times 10^{-11}\,e^{(-1300/T)},\ T \leq 400$ K	Baulch *et al.* (1994)
		$F_c = 0.44$	Baulch *et al.* (1994)
R154	$\mathrm{H + C_2H_3 \longrightarrow C_2H_2 + H_2}$	$1.5 \times 10^{-12}\,T^{0.5}$	Moses *et al.* (2005)
R156	$\mathrm{H + C_2H_3} \xrightarrow{\mathrm{M}} \mathrm{C_2H_4}$	$k_0 = 2.3 \times 10^{-24}\,T^{-1}$	Moses *et al.* (2005)
		$k_\infty = 1.8 \times 10^{-10}$	Moses *et al.* (2005)
R158	$\mathrm{H + C_2H_4} \xrightarrow{\mathrm{M}} \mathrm{C_2H_5}$	$k_0 = 1.3 \times 10^{-29}\,e^{(-380/T)},\ T > 302.4$ K	Baulch *et al.* (1994)
		$k_0 = 3.7 \times 10^{-30},\ T \leq 302.4$ K	Moses *et al.* (2005)
		$k_\infty = 6.6 \times 10^{-15}\,T^{1.28}\,e^{(-650/T)}$	Baulch *et al.* (1994)
		$F_c = 0.24e^{(-T/40)}\ +\ 0.76e^{(-T/1025)}$	Baulch *et al.* (1994)
R160	$\mathrm{H + C_2H_5 \longrightarrow 2\,CH_3}$	1.25×10^{-10}	Sillesen *et al.* (1993)
R162	$\mathrm{H + C_2H_5 \longrightarrow C_2H_4 + H_2}$	3.0×10^{-12}	Tsang and Hampson (1986)
R164	$\mathrm{H + C_2H_5} \xrightarrow{\mathrm{M}} \mathrm{C_2H_6}$	$k_0 = 4.0 \times 10^{-19}\,T^{-3}\,e^{(-600/T)}, T > 200$ K	Est., Moses *et al.* (2005)
		$k_0 = 2.489 \times 10^{-27},\ T \leq 200$ K	Est., Moses *et al.* (2005)
		$k_\infty = 2.0 \times 10^{-10}$	based on Sillesen *et al.* (1993)
R166	$\mathrm{H + C_2H_6 \longrightarrow C_2H_5 + H_2}$	$2.35 \times 10^{-15}\,T^{1.5}\,e^{(-3725/T)}$	Baulch *et al.* (1992)
R168	$\mathrm{H + C_3H_2} \xrightarrow{\mathrm{M}} \mathrm{C_3H_3}$	$k_0 = 4.0 \times 10^{-19}\,T^{-3}\,e^{(-600/T)}$	Est. based on H + $\mathrm{C_2H_5}$
		$k_0 = 2.489 \times 10^{-27},\ T \leq 200$ K	Estimate
		$k_\infty = 2.0 \times 10^{-10}$	Estimate
R170	$\mathrm{H + C_3H_3} \xrightarrow{\mathrm{M}} \mathrm{CH_3C_2H}$	$k_0 = 9.4 \times 10^{-20}\,T^{-3.3},\ T > 140$ K	Est., Moses
		$k_0 = 7.779 \times 10^{-27},\ T \leq 140$ K	*et al.* (2005)
		$k_\infty = 1.0 \times 10^{-10}$	Estimate
R172	$\mathrm{H + C_3H_3} \xrightarrow{\mathrm{M}} \mathrm{CH_2CCH_2}$	$k_0 = 1.6 \times 10^{-20}\,T^{-3.3}$	Est., Moses
		$k_0 = 1.324 \times 10^{-27},\ T \leq 140$ K	*et al.* (2005)
		$k_\infty = 1.0 \times 10^{-10}$	Estimate
R174	$\mathrm{H + CH_3C_2H \longrightarrow C_3H_3 + H_2}$	$4.7 \times 10^{-16}\,T^{1.74}\,e^{(-3873/T)}$	Wang *et al.* (2000)
R176	$\mathrm{H + CH_3C_2H \longrightarrow C_2H_2 + CH_3}$	$2.5 \times 10^{-10}\,e^{(-3052/T)}$	based on Wang *et al.* (2000)
R178	$\mathrm{H + CH_3C_2H} \xrightarrow{\mathrm{M}} \mathrm{C_3H_5}$	$k_0 = 3.2 \times 10^{-24}\,T^{-1.7}\,e^{(-300/T)}$	based on falloff data
		$k_\infty = 2.93 \times 10^{-18}\,T^{2.496}\,e^{(-470.25/T)},\ T \leq 622$ K	of Whytock *et al.* (1976b),
		$k_\infty = 9.4 \times 10^{-11}\,e^{(-1234/T)},\ 622 < T \leq 1200$ K	see also
		$k_\infty = 5.0 \times 10^{-11}\,e^{(-1180/T)},\ T > 1200$ K	Wang *et al.* (2000)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R180	$H + CH_2CCH_2 \longrightarrow CH_3C_2H + H$	$3.0 \times 10^{-18}\, T^2\, e^{(-52/T)}$	Est. based on Wagner and Zellner
R182	$H + CH_2CCH_2 \xrightarrow{M} C_3H_5$	$k_0 = 3.0 \times 10^{-28}\, e^{(-360/T)}$	(1972b) and Aleksandrov *et al.*
		$k_\infty = 1.0 \times 10^{-18}\, T^2\, e^{(-52/T)}$	(1980)
R184	$H + C_3H_5 \longrightarrow CH_3C_2H + H_2$	1.5×10^{-11}	Est. based on Tsang (1991),
R186	$H + C_3H_5 \longrightarrow CH_2CCH_2 + H_2$	1.5×10^{-11}	Hanning-Lee and Pilling (1992),
R188	$H + C_3H_5 \xrightarrow{M} C_3H_6$	$k_0 = 3.2 \times 10^{-24}\, T^{-1.7}\, e^{(-300/T)}$	and Harding *et al.*
		$k_\infty = 9.69 \times 10^{-11}\, T^{0.18}\, e^{(63/T)}$	(2007)
R190	$H + C_3H_6 \longrightarrow C_3H_5 + H_2$	$2.87 \times 10^{-19}\, T^{2.5}\, e^{(-1254/T)}$	Tsang (1991)
R192	$H + C_3H_6 \longrightarrow CH_3 + C_2H_4$	$2.2 \times 10^{-11}\, e^{(-1641/T)}$	Tsang (1991)
R194	$H + C_3H_6 \xrightarrow{M} C_3H_7$	$k_0 = 1.3 \times 10^{-28}\, e^{(-380/T)}$	Est. 10 × k_0 for H + C_2H_4
		$k_\infty = 9.47 \times 10^{-15}\, T^{1.16}\, e^{(-440/T)}$	Seakins *et al.* (1993)
R196	$H + C_3H_7 \longrightarrow C_3H_6 + H_2$	3.0×10^{-12}	Tsang (1988)
R198	$H + C_3H_7 \longrightarrow C_2H_5 + CH_3$	6.0×10^{-11}	Est. based on Tsang (1988)
R200	$H + C_3H_7 \xrightarrow{M} C_3H_8$	$k_0 = 4.0 \times 10^{-18}\, T^{-3.0}\, e^{(-600/T)}$	Est., 10× H + C_2H_5
		$k_0 = 2.489 \times 10^{-26},\ T \leq 200$ K	
		$k_\infty = 2.49 \times 10^{-10}$	Munk *et al.* (1986)
R202	$H + C_3H_8 \longrightarrow C_3H_7 + H_2$	$2.16 \times 10^{-18}\, T^{2.4}\, e^{(-2250/T)}$	Tsang (1988)
R204	$H + C_4H \xrightarrow{M} C_4H_2$	$k_0 = 1.26 \times 10^{-18}\, T^{-3.1}\, e^{(-721/T)}$	Est. based on H + C_2H
		$k_\infty = 3.0 \times 10^{-10}$	Est. based on H + C_2H
R206	$H + C_4H_2 \xrightarrow{M} C_4H_3$	$k_0 = 2.0 \times 10^{-26}\, e^{(-740/T)}$	Est. based on Nava *et al.* (1986)
		$k_\infty = 1.5 \times 10^{-10}\, e^{(-1184/T)}$	Est. based on Nava *et al.* (1986)
R208	$H + C_4H_3 \longrightarrow 2\,C_2H_2$	1.0×10^{-11}	Estimate
R210	$H + C_4H_3 \longrightarrow C_4H_2 + H_2$	6.0×10^{-11}	Estimate
R212	$H + C_4H_3 \xrightarrow{M} C_4H_4$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 8.56 \times 10^{-10}\, e^{(-405/T)}$	Durán *et al.* (1988)
R214	$H + C_4H_4 \xrightarrow{M} C_4H_5$	$k_0 = 1.5 \times 10^{-20}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 9.35 \times 10^{-12}\, e^{(-309/T)}$	Est. from Ancia *et al.* (1996) and Schwanebeck and Warnatz (1975)
R216	$H + C_4H_5 \longrightarrow C_4H_4 + H_2$	2.0×10^{-11}	Estimate
R218	$H + C_4H_5 \xrightarrow{M} C_4H_6$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 1.0 \times 10^{-10}$	Gladstone *et al.* (1996)
R220	$H + C_4H_6 \longrightarrow C_4H_5 + H_2$	$1.05 \times 10^{-13}\, T^{0.7}\, e^{(-3019/T)}$	Weissman and Bensen (1988)
R222	$H + C_4H_8 \longrightarrow C_3H_6 + CH_3$	$2.86 \times 10^{-11}\, e^{(-1810/T)}$	Tsang and Walker (1989)
R224	$H + C_4H_8 \xrightarrow{M} C_4H_9$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 1.39 \times 10^{-12}\, T^{0.5}\, e^{(-478/T)}$	literature review
R226	$H + C_4H_9 \longrightarrow C_4H_8 + H_2$	1.5×10^{-12}	Tsang (1990)
R228	$H + C_4H_9 \xrightarrow{M} C_4H_{10}$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 5.9 \times 10^{-11}$	Tsang (1990)
R230	$H + C_4H_{10} \longrightarrow C_4H_9 + H_2$	$3.49 \times 10^{-11}\, e^{(-3969/T)}$	Nicholas and Vaghijiani (1989)
R232	$H + C_5H_3 \longrightarrow C_3H_2 + C_2H_2$	$1.5 \times 10^{-12}\, T^{0.5}$	Est. based on H + C_2H_3
R234	$H + C_6H_2 \xrightarrow{M} C_6H_3$	$k_0 = 1.0 \times 10^{-29}\, e^{(-360/T)}$	Estimate
		$k_\infty = 1.5 \times 10^{-10}\, e^{(-1184/T)}$	Est. based on H + C_4H_2
R236	$H + C_6H_3 \longrightarrow C_6H_2 + H_2$	$3.0 \times 10^{-12}\, T^{0.5}$	Est. based on H + C_2H_3
R238	$H + C_6H_3 \longrightarrow C_4H_2 + C_2H_2$	$3.99 \times 10^{-5}\, T^{-1.6}\, e^{(-1409/T)}$	Wang and Frenklach (1997)
R240	$H + C_6H_3 \xrightarrow{M} C_6H_4$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 8.56 \times 10^{-10}\, e^{(-405/T)}$	Est. based on H + C_4H_3

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R242	$H + C_6H_4 \xrightarrow{M} C_6H_5$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 9.35 \times 10^{-12}\, e^{(-309/T)}$	Est. based on H + C_4H_4
R244	$H + C_6H_5 \xrightarrow{M} C_6H_6$	$k_0 = 1.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
		$k_\infty = 3.65 \times 10^{-10}$	Davis *et al.* (1996)
R246	$C + H_2 \xrightarrow{M} {}^3CH_2$	$k_0 = 7.0 \times 10^{-32}$	Husain and Young (1975)
		$k_\infty = 2.06 \times 10^{-11}\, e^{(-57/T)}$	Harding *et al.* (1993)
R248	$C + C_2H_2 \longrightarrow C_3H_2$	5.95×10^{-10}	Liao and Herbst (1995)
R250	$C + C_2H_4 \longrightarrow C_3H_3 + H$	2.0×10^{-10}	Haider and Husain (1993)
R252	$CH + H_2 \longrightarrow H + {}^3CH_2$	$3.75 \times 10^{-10}\, e^{(-1663/T)}$	Becker *et al.* (1991)
R254	$CH + H_2 \xrightarrow{M} CH_3$	$k_0 = 9.0 \times 10^{-31}\, e^{(550/T)}$	Fit to data from Becker *et al.* (1991) and Berman and Lin (1984)
		$k_\infty = 8.55 \times 10^{-11}\, T^{0.15}$	Fulle and Hippler (1997)
R256	$CH + {}^3CH_2 \longrightarrow C_2H_2 + H$	$3.96 \times 10^{-8}\, T^{-1.04}\, e^{(-36.1/T)}$	Est. based on CH + CH_4
		$1.58 \times 10^{-8}\, T^{-0.9}$	for $T > 295$ K
R258	$CH + CH_3 \longrightarrow C_2H_3 + H$	$3.96 \times 10^{-8}\, T^{-1.04}\, e^{(-36.1/T)}$	Est. based on CH + CH_4
		$1.58 \times 10^{-8}\, T^{-0.9}$	for $T > 295$ K
R260	$CH + CH_4 \longrightarrow C_2H_4 + H$	$3.96 \times 10^{-8}\, T^{-1.04}\, e^{(-36.1/T)}$	Canosa *et al.* (1997)
		$1.58 \times 10^{-8}\, T^{-0.9}, T > 295$ K	Est. based on Blitz *et al.* (1997)
R262	$CH + C_2H_2 \longrightarrow C_3H_2 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Canosa *et al.* (1997)
		$3.255 \times 10^{-10}\, e^{(61/T)}, T > 295$ K	Est. based on Berman *et al.* (1982)
R264	$CH + C_2H_3 \longrightarrow C_3H_3 + H$	$7.74 \times 10^{-9}\, T^{-0.55}\, e^{(-29.6/T)}$	Est. based on CH + C_2H_4
		$1.746 \times 10^{-10}\, e^{(173/T)}$	for $T > 295$ K
R266	$CH + C_2H_4 \longrightarrow CH_2CCH_2 + H$	$7.74 \times 10^{-9}\, T^{-0.55}\, e^{(-29.6/T)}$	Canosa *et al.* (1997)
		$1.746 \times 10^{-10}\, e^{(173/T)}, T > 295$ K	Est. based on Berman *et al.* (1982)
R268	$CH + C_2H_5 \longrightarrow C_3H_5 + H$	$3.8 \times 10^{-8}\, T^{-0.86}\, e^{(-33.5/T)}$	Est. based on CH + C_2H_6
		$1.64 \times 10^{-10}\, e^{(132/T)}$	for $T > 295$ K
R270	$CH + C_2H_6 \longrightarrow C_3H_6 + H$	$3.8 \times 10^{-8}\, T^{-0.86}\, e^{(-33.5/T)}$	Canosa *et al.* (1997)
		$1.64 \times 10^{-10}\, e^{(132/T)}, T > 295$ K	Est. based on Berman and Lin (1983)
R272	$CH + C_3H_2 \longrightarrow C_4H_2 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R274	$CH + C_3H_3 \longrightarrow C_4H_3 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R276	$CH + CH_3C_2H \longrightarrow C_4H_4 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R278	$CH + CH_2CCH_2 \longrightarrow C_4H_4 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R280	$CH + C_5H_3 \longrightarrow C_6H_3 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R282	$CH + C_5H_4 \longrightarrow C_6H_4 + H$	$1.59 \times 10^{-9}\, T^{-0.23}\, e^{(-16/T)}$	Est. based on CH + C_2H_2
		$3.255 \times 10^{-10}\, e^{(61/T)}$	for $T > 295$ K
R284	${}^1CH_2 + H_2 \longrightarrow {}^3CH_2 + H_2$	1.26×10^{-11}	Braun *et al.* (1970); and
R286	${}^1CH_2 + H_2 \longrightarrow CH_3 + H$	9.24×10^{-11}	Langford *et al.* (1983)
R288	${}^1CH_2 + CH_3 \longrightarrow C_2H_4 + H$	7.0×10^{-11}	Est. based on ${}^3CH_2$ + CH_3
R290	${}^1CH_2 + CH_4 \longrightarrow {}^3CH_2 + CH_4$	1.2×10^{-11}	Böhland *et al.* (1985b)
R292	${}^1CH_2 + CH_4 \longrightarrow 2\,CH_3$	5.9×10^{-11}	Böhland *et al.* (1985b)

TABLE S1 (*cont.*)

Reaction				Rate constant	Reference
R294	$^1CH_2 + C_2H_2$	$\longrightarrow$	$^3CH_2 + C_2H_2$	6.6×10^{-11}	Böhland *et al.* (1986)
R296	$^1CH_2 + C_2H_2$	$\xrightarrow{M}$	CH_3C_2H	$k_0 = 7.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 2.5 \times 10^{-11}$	Based on Böhland *et al.* (1986)
R298	$^1CH_2 + C_2H_2$	$\xrightarrow{M}$	CH_2CCH_2	$k_0 = 7.5 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 2.5 \times 10^{-11}$	Based on Böhland *et al.* (1986)
R300	$^1CH_2 + C_2H_2$	$\longrightarrow$	$C_3H_3 + H$	2.5×10^{-10}	Based on Böhland *et al.* (1986)
R302	$^1CH_2 + C_2H_4$	$\longrightarrow$	$^3CH_2 + C_2H_4$	1.8×10^{-11}	Böhland *et al.* (1985b)
R304	$^1CH_2 + C_2H_4$	$\xrightarrow{M}$	C_3H_6	$k_0 = 1.5 \times 10^{-18}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 1.32 \times 10^{-10}$	Böhland *et al.* (1985b)
R306	$^1CH_2 + C_2H_6$	$\longrightarrow$	$^3CH_2 + C_2H_6$	3.6×10^{-11}	Baulch *et al.* (1992)
R308	$^3CH_2 + H_2$	$\longrightarrow$	$H + CH_3$	$8.3 \times 10^{-19}\, T^2\, e^{(-3638/T)}$	Wang and Frenklach (1997)
R310	$2\,^3CH_2$	$\longrightarrow$	$C_2H_2 + 2\,H$	$1.8 \times 10^{-10}\, e^{(-400/T)}$	Baulch *et al.* (1992)
R312	$^3CH_2 + CH_3$	$\longrightarrow$	$C_2H_4 + H$	7.0×10^{-11}	Baulch *et al.* (1992)
R314	$^3CH_2 + CH_4$	$\longrightarrow$	$2\,CH_3$	$7.1 \times 10^{-12}\, e^{(-5051/T)}$	Böhland *et al.* (1985a)
R316	$^3CH_2 + C_2H$	$\longrightarrow$	$C_3H_2 + H$	3.0×10^{-11}	Estimate
R318	$^3CH_2 + C_2H$	$\longrightarrow$	$C_2H_2 + CH$	3.0×10^{-11}	Estimate
R320	$^3CH_2 + C_2H_2$	$\longrightarrow$	$C_3H_2 + H_2$	$1.0 \times 10^{-12}\, e^{(-3332/T)}$	Böhland *et al.* (1986)
R322	$^3CH_2 + C_2H_2$	$\longrightarrow$	$C_3H_3 + H$	$1.9 \times 10^{-11}\, e^{(-3332/T)}$	Böhland *et al.* (1986)
R324	$^3CH_2 + C_2H_2$	$\xrightarrow{M}$	CH_3C_2H	$k_0 = 1.5 \times 10^{-18}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 2.0 \times 10^{-12}\, e^{(-3332/T)}$	Est. based on Böhland *et al.* (1986)
R326	$^3CH_2 + C_2H_3$	$\longrightarrow$	$C_2H_2 + CH_3$	8.0×10^{-11}	Estimate
R328	$^3CH_2 + C_2H_4$	$\longrightarrow$	$C_3H_5 + H$	$4.25 \times 10^{-12}\, e^{(-2658/T)}$	Est. based on Kraus *et al.* (1993)
R330	$^3CH_2 + C_2H_4$	$\xrightarrow{M}$	C_3H_6	$k_0 = 1.5 \times 10^{-18}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 1.0 \times 10^{-12}\, e^{(-2658/T)}$	Est. based on Kraus *et al.* (1993)
R332	$^3CH_2 + C_2H_5$	$\longrightarrow$	$C_2H_4 + CH_3$	8.0×10^{-11}	Estimate
R334	$^3CH_2 + C_2H_6$	$\longrightarrow$	$C_2H_5 + CH_3$	$1.07 \times 10^{-11}\, e^{(-3981/T)}$	Böhland *et al.* (1985a)
R336	$^3CH_2 + C_3H_2$	$\longrightarrow$	$C_4H_3 + H$	8.0×10^{-11}	Estimate
R338	$^3CH_2 + C_3H_3$	$\longrightarrow$	$C_4H_4 + H$	3.0×10^{-11}	Estimate
R340	$^3CH_2 + C_3H_5$	$\longrightarrow$	$C_4H_6 + H$	5.0×10^{-11}	Tsang (1991)
R342	$^3CH_2 + C_3H_6$	$\longrightarrow$	$C_3H_5 + CH_3$	$1.2 \times 10^{-12}\, e^{(-3116/T)}$	Tsang (1991)
R344	$^3CH_2 + C_4H_2$	$\longrightarrow$	$C_5H_3 + H$	$2.16 \times 10^{-11}\, e^{(-2165/T)}$	Böhland *et al.* (1986)
R346	$^3CH_2 + C_5H_3$	$\longrightarrow$	$C_6H_4 + H$	1.0×10^{-11}	Estimate
R348	$^3CH_2 + C_5H_4$	$\longrightarrow$	$C_6H_5 + H$	$1.0 \times 10^{-11}\, e^{(-3000/T)}$	Estimate
R350	$2\,CH_3$	$\longrightarrow$	$C_2H_4 + H_2$	$1.66 \times 10^{-10}\, e^{(-16103/T)}$	Hidaka *et al.* (1990)
R352	$2\,CH_3$	$\xrightarrow{M}$	C_2H_6	$k_0 = 3.51 \times 10^{-7}\, T^{-7.03}\, e^{(-1390/T)}$	for $T > 300$ K, based on Slagle *et al.* (1988)
				$k_0 = 6.15 \times 10^{-18}\, T^{-3.5}$	for $T \leq 300$K, Estimate
				$k_\infty = 1.12 \times 10^{-9}\, T^{-0.5}\, e^{(-25/T)}$	Data compilation; fits
				$F_c = 0.62e^{(-T/1180)} + 0.38e^{(-T/73)}$	falloff region data
					of Walter *et al.* (1990)
R354	$CH_3 + C_2H$	$\xrightarrow{M}$	CH_3C_2H	$k_0 = 1.5 \times 10^{-18}\, T^{-3}\, e^{(-300/T)}$	Estimate
				$k_\infty = 1.7 \times 10^{-10}$	Est., Fahr *et al.* (2001)
R356	$CH_3 + C_2H_2$	$\longrightarrow$	$CH_2CCH_2 + H$	$1.12 \times 10^{-4}\, T^{-2.08}\, e^{(-15898/T)}$	Dean and Westmoreland (1987)
R358	$CH_3 + C_2H_2$	$\longrightarrow$	C_3H_5	$3.81 \times 10^{6}\, T^{-5.98}\, e^{(-6709/T)}$	for $T \leq 1000$ K, Diau and Lin (1994)
				$6.39 \times 10^{32}\, T^{-13.7}\, e^{(-14036/T)}$	for $T > 1000$ K, Diau and Lin (1994)

TABLE S1 (*cont.*)

Reaction	Rate constant	Reference
R360 $CH_3 + C_2H_3 \longrightarrow CH_4 + C_2H_2$	$1.5 \times 10^{-11}\, e^{(385/T)}$	Based on
	5.413×10^{-11} for $T \leq 300$ K	Stoliarov *et al.* (2000),
R362 $CH_3 + C_2H_3 \longrightarrow C_3H_5 + H$	$k_{\infty,364} - k_{364}[M]$	Fahr *et al.* (1991),
R364 $CH_3 + C_2H_3 \xrightarrow{M} C_3H_6$	$k_0 = 2.2 \times 10^{-23}\, T^{-1.26}\, e^{(-362/T)}$	Fahr *et al.* (1999), and
	$k_\infty = 3.3 \times 10^{-11}\, e^{(236/T)}$	Thorn *et al.* (2000)
	$k_\infty = 7.25 \times 10^{-11}$ for $T < 300$ K	
R366 $CH_3 + C_2H_4 \longrightarrow C_2H_3 + CH_4$	$3.77 \times 10^{-19}\, T^2\, e^{(-4629/T)}$	Wang and Frenklach (1997)
R368 $CH_3 + C_2H_4 \longrightarrow C_3H_7$	$3.5 \times 10^{-13}\, e^{(-3700/T)}$	Baulch *et al.* (1992)
R370 $CH_3 + C_2H_5 \longrightarrow C_2H_4 + CH_4$	2.0×10^{-12}	Baulch *et al.* (1992)
R372 $CH_3 + C_2H_5 \xrightarrow{M} C_3H_8$	$k_0 = 7.5 \times 10^{-17}\, T^{-3}\, e^{(-300/T)}$	Estimate
	$k_\infty = 6.64 \times 10^{-11}$	Sillesen *et al.* (1993)
R374 $CH_3 + C_2H_6 \longrightarrow C_2H_5 + CH_4$	$2.5 \times 10^{-31}\, T^6\, e^{(-3043/T)}$	Baulch *et al.* (1992)
R376 $CH_3 + C_3H_2 \longrightarrow C_2H_2 + C_2H_3$	7.0×10^{-11}	Estimate
R378 $CH_3 + C_3H_2 \longrightarrow C_4H_4 + H$	1.0×10^{-11}	Estimate
R380 $CH_3 + C_3H_3 \xrightarrow{M} C_4H_6$	$k_0 = 1.5 \times 10^{-18}\, T^{-3}\, e^{(-300/T)}$	Estimate
	$k_\infty = 1.66 \times 10^{-11}$	Est. based on Wu and Kern (1987)
R382 $CH_3 + C_3H_5 \longrightarrow CH_4 + CH_3C_2H$	2.0×10^{-11}	Estimate based on
R384 $CH_3 + C_3H_5 \longrightarrow CH_4 + CH_2CCH_2$	2.0×10^{-11}	Garland and Bayes (1990) and
R386 $CH_3 + C_3H_5 \xrightarrow{M} C_4H_8$	$k_0 = 3.7 \times 10^{-19}\, T^{-3}\, e^{(-300/T)}$	Knyazev and Slagle (2001)
	$k_\infty = 5.0 \times 10^{-11}$	
R388 $CH_3 + C_3H_6 \longrightarrow CH_4 + C_3H_5$	$2.32 \times 10^{-13}\, e^{(-4390/T)}$	Kinsman and Roscoe (1994)
R390 $CH_3 + C_3H_7 \longrightarrow CH_4 + C_3H_6$	$1.9 \times 10^{-11}\, T^{-0.3}$	Tsang (1988)
R392 $CH_3 + C_3H_7 \xrightarrow{M} C_4H_{10}$	$k_0 = 7.07 \times 10^{-22}\, e^{(255/T)},\ T \leq 200$ K	Est. 0.01 × Laufer *et al.* (1983)
	$k_0 = 4.46 \times 10^{-26}\, e^{(2189/T)},\ T > 200$ K	Est. 0.01 × Laufer *et al.* (1983)
	$k_\infty = 3.2 \times 10^{-10}\, T^{-0.32}$	Tsang (1988)
R394 $CH_3 + C_3H_8 \longrightarrow CH_4 + C_3H_7$	$1.5 \times 10^{-24}\, T^{3.7}\, e^{(-3600/T)}$	Tsang (1988)
R396 $CH_3 + C_4H_5 \longrightarrow CH_4 + C_4H_4$	3.3×10^{-11}	Estimate
R398 $CH_3 + C_5H_3 \longrightarrow C_6H_5 + H$	2.0×10^{-13}	Estimate
R400 $CH_3 + C_5H_3 \longrightarrow C_4H_4 + C_2H_2$	1.0×10^{-11}	Estimate
R402 $CH_3 + C_5H_3 \longrightarrow 2\,C_3H_3$	4.0×10^{-11}	Estimate
R404 $CH_3 + C_5H_3 \xrightarrow{M} C_6H_6$	$k_0 = 1.5 \times 10^{-17}\, T^{-3}\, e^{(-300/T)}$	Estimate
	$k_\infty = 3.2 \times 10^{-10}\, T^{-0.32}$	Est. based on $CH_3 + C_3H_7$
R406 $C_2 + H_2 \longrightarrow C_2H + H$	$1.77 \times 10^{-10}\, e^{(-1469/T)}$	Pitts *et al.* (1982)
R408 $C_2 + CH_4 \longrightarrow C_2H + CH_3$	$5.05 \times 10^{-11}\, e^{(-297/T)}$	Pitts *et al.* (1982)
R410 $C_2H + H_2 \longrightarrow C_2H_2 + H$	$9.2 \times 10^{-18}\, T^{2.17}\, e^{(-478/T)}$	Opansky and Leone (1996b)
R412 $C_2H + CH_4 \longrightarrow C_2H_2 + CH_3$	$1.2 \times 10^{-11}\, e^{(-491/T)}$	Opansky and Leone (1996a)
R414 $2\,C_2H \xrightarrow{M} C_4H_2$	$k_0 = 1.5 \times 10^{-20}\, T^{-3}\, e^{(-300/T)}$	Estimate
	$k_\infty = 2.6 \times 10^{-10}$	Est., Fahr *et al.* (2001)
R416 $C_2H + C_2H_2 \longrightarrow C_4H_2 + H$	1.3×10^{-10}	Vakhtin *et al.* (2001)
R418 $C_2H + C_2H_3 \longrightarrow C_4H_3 + H$	$1.45 \times 10^{-10}\, e^{(134/T)}$	Est. based on Durán *et al.* (1988)
R420 $C_2H + C_2H_3 \longrightarrow 2\,C_2H_2$	1.6×10^{-12}	Tsang and Hampson (1986)
R422 $C_2H + C_2H_4 \longrightarrow C_4H_4 + H$	$7.8 \times 10^{-11}\, e^{(134/T)}$	Opansky and Leone (1996b)
R424 $C_2H + C_2H_5 \longrightarrow C_4H_5 + H$	1.2×10^{-10}	Est. based on Fahr *et al.* (2001)
R426 $C_2H + C_2H_5 \longrightarrow C_3H_3 + CH_3$	3.0×10^{-11}	Tsang and Hampson (1986)
R428 $C_2H + C_2H_5 \longrightarrow C_2H_2 + C_2H_4$	3.0×10^{-12}	Tsang and Hampson (1986)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R430	$C_2H + C_2H_6 \longrightarrow C_2H_2 + C_2H_5$	$1.19 \times 10^{-12}\,T^{0.54}\,e^{(180/T)}$	Ceursters *et al.* (2001)
R432	$C_2H + CH_3C_2H \longrightarrow C_2H_2 + C_3H_3$	1.66×10^{-11}	Wu and Kern (1987)
R434	$C_2H + CH_2CCH_2 \longrightarrow C_2H_2 + C_3H_3$	1.66×10^{-11}	Wu and Kern (1987)
R436	$C_2H + C_3H_8 \longrightarrow C_2H_2 + C_3H_7$	$7.8 \times 10^{-11}\,e^{(3/T)}$	Hoobler *et al.* (1997)
R438	$C_2H + C_4H_2 \longrightarrow C_6H_2 + H$	1.3×10^{-10}	Est. based on C_2H + C_2H_2
R440	$C_2H_2 + C_4H_4 \longrightarrow C_6H_6$	$7.4 \times 10^{-13}\,e^{(-15143/T)}$	Chanmugathas and Heicklen (1986)
R442	$C_2H_3 + H_2 \longrightarrow C_2H_4 + H$	$1.57 \times 10^{-20}\,T^{2.56}\,e^{(-2529/T)}$	Knyazev *et al.* (1996a)
R444	$C_2H_3 + C_2H_2 \longrightarrow C_4H_4 + H$	$3.2 \times 10^{-12}\,e^{(-3020/T)}$	Knyazev *et al.* (1996b)
R446	$C_2H_3 + C_2H_2 \xrightarrow{M} C_4H_5$	$k_0 = 8.2 \times 10^{-30}\,e^{(-352/T)}$	Estimate
		$k_\infty = 4.17 \times 10^{-19}\,T^{1.9}\,e^{(-1058/T)}$	Weissman and Benson (1988)
R448	$2\,C_2H_3 \longrightarrow CH_3 + C_3H_3$	2.5×10^{-11}	Ismail *et al.* (2009)
R450	$2\,C_2H_3 \longrightarrow C_2H_4 + C_2H_2$	1.5×10^{-11}	Ismail *et al.* (2009)
R452	$2\,C_2H_3 \xrightarrow{M} C_4H_6$	$k_0 = 5.0 \times 10^{-18}\,T^{-3.75}\,e^{(-300/T)}$	Estimate
		$k_\infty = 1.4 \times 10^{-10}$	Fahr *et al.* (1991)
R454	$C_2H_3 + C_2H_4 \longrightarrow C_4H_6 + H$	$1.05 \times 10^{-12}\,e^{(-1559/T)}$	Fahr and Stein (1988)
R456	$C_2H_3 + C_2H_5 \longrightarrow 2\,C_2H_4$	8.0×10^{-13}	Tsang and Hampson (1986)
R458	$C_2H_3 + C_2H_5 \longrightarrow C_2H_6 + C_2H_2$	8.0×10^{-13}	Tsang and Hampson (1986)
R460	$C_2H_3 + C_2H_5 \longrightarrow CH_3 + C_3H_5$	$2.5 \times 10^{-11} A/(1 + A),$ where	Tsang and Hampson (1986)
		$A = 2.5 \times 10^{-36}\,T^{11.25}\,e^{(-300/T)}$	
R462	$C_2H_3 + C_2H_5 \xrightarrow{M} C_4H_8$	$k_0 = 7.5 \times 10^{-17}\,T^{-3}\,e^{(-300/T)}$	Estimate
		$k_\infty = 2.5 \times 10^{-11} - k_{460}$	Tsang and Hampson (1986)
R464	$C_2H_3 + C_2H_6 \longrightarrow C_2H_4 + C_2H_5$	$9.98 \times 10^{-22}\,T^{3.3}\,e^{(-5285/T)}$	Tsang and Hampson (1986)
R466	$C_2H_3 + C_3H_2 \longrightarrow C_3H_3 + C_2H_2$	8.0×10^{-11}	Estimate
R468	$C_2H_3 + C_3H_3 \longrightarrow CH_3C_2H + C_2H_2$	2.0×10^{-11}	Estimate
R470	$C_2H_3 + C_3H_5 \longrightarrow CH_2CCH_2 + C_2H_4$	2.0×10^{-11}	Estimate
R472	$C_2H_3 + C_3H_5 \longrightarrow C_3H_6 + C_2H_2$	2.0×10^{-11}	Estimate
R474	$C_2H_3 + C_3H_8 \longrightarrow C_2H_4 + C_3H_7$	$1.7 \times 10^{-21}\,T^{3.1}\,e^{(-4443/T)}$	Tsang (1988)
R476	$C_2H_3 + C_4H_3 \longrightarrow C_6H_5 + H$	9.963×10^{-12}	Pope and Miller (2000)
R478	$C_2H_4 \xrightarrow{M} C_2H_2 + H_2$	$k_0 = 5.8 \times 10^{-8}\,e^{(-36000/T)}$	Baulch *et al.* (1994)
		$k_\infty = 3.98 \times 10^{15}\,e^{(-51056/T)}$	Melius *et al.* (1992)
R480	$C_2H_5 + C_2H_4 \longrightarrow C_4H_9$	$1.8 \times 10^{-13}\,e^{(-3670/T)}$	Baulch *et al.* (1994)
R482	$2\,C_2H_5 \longrightarrow C_2H_6 + C_2H_4$	2.4×10^{-12}	Baulch *et al.* (1992)
R484	$2\,C_2H_5 \xrightarrow{M} C_4H_{10}$	$k_0 = 1.5 \times 10^{-17}\,T^{-3}\,e^{(-300/T)}$	Estimate
		$k_\infty = 1.4 \times 10^{-11}\,e^{(35/T)}$	Gladstone *et al.* (1996)
R486	$C_2H_5 + C_3H_2 \longrightarrow C_3H_3 + C_2H_4$	8.0×10^{-11}	Estimate
R488	$C_2H_5 + C_3H_3 \longrightarrow CH_3C_2H + C_2H_4$	2.0×10^{-11}	Estimate
R490	$C_2H_5 + C_3H_5 \longrightarrow CH_2CCH_2 + C_2H_6$	$1.6 \times 10^{-12}\,e^{(66/T)}$	Tsang (1991)
R492	$C_2H_5 + C_3H_5 \longrightarrow C_3H_6 + C_2H_4$	$4.3 \times 10^{-12}\,e^{(66/T)}$	Tsang (1991)
R494	$C_2H_5 + C_4H_3 \longrightarrow C_4H_2 + C_2H_6$	8.0×10^{-11}	Estimate
R496	$C_3H_2 + C_2H_2 \xrightarrow{M} C_5H_4$	$k_0 = 1.5 \times 10^{-18}\,T^{-3}\,e^{(-300/T)}$	Estimate
		$k_\infty = 2.0 \times 10^{-11}\,e^{(-3330/T)}$	Est. based on $^3CH_2$ + C_2H_2
R498	$C_3H_2 + C_3H_3 \longrightarrow C_6H_4 + H$	$8.0 \times 10^{-9}\,T^{-1}$	Estimate
R500	$2\,C_3H_3 \xrightarrow{M} l\text{-}C_6H_6$	$k_0 = 2.81 \times 10^{-21}\,T^{-2}$	Based on Fahr and Nayak (2000)
		$k_\infty = 4.1 \times 10^{-11}$	and Atkinson and Hudgens (1999)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R502	$2\,C_3H_3 \xrightarrow{M} C_6H_6$	$k_0 = 1.48 \times 10^{-22}\,T^{-2}$	Based on Fahr and Nayak (2000)
		$k_\infty = 6.0 \times 10^{-12}$	and Atkinson and Hudgens (1999)
R504	$C_3H_3 + C_3H_5 \longrightarrow 2\,CH_3C_2H$	2.0×10^{-11}	Estimate
R506	$2\,C_3H_5 \longrightarrow CH_2CCH_2 + C_3H_6$	$1.4 \times 10^{-13}\,e^{(132/T)}$	Tsang (1991)
R508	$C_3H_5 + C_3H_7 \longrightarrow C_3H_8 + CH_2CCH_2$	$7.6 \times 10^{-12}\,T^{-0.35}\,e^{(66/T)}$	Tsang (1991)
R510	$C_4H + H_2 \longrightarrow H + C_4H_2$	$9.2 \times 10^{-18}\,T^{2.17}\,e^{(-478/T)}$	Est. based on $C_2H + H_2$
R512	$C_4H + CH_4 \longrightarrow C_4H_2 + CH_3$	$1.2 \times 10^{-11}\,e^{(-491/T)}$	Est. based on $C_2H + CH_4$
R514	$C_4H + C_2H_2 \longrightarrow C_6H_2 + H$	2.5×10^{-11}	Brachhold *et al.* (1988)
R516	$C_4H + C_2H_6 \longrightarrow C_4H_2 + C_2H_5$	$1.19 \times 10^{-12}\,T^{0.54}\,e^{(180/T)}$	Est. based on $C_2H + C_2H_6$
R518	$C_4H_3 + C_2H_2 \xrightarrow{M} C_6H_5$	$k_0 = 3.0 \times 10^{-23}\,T^{-1}$	Estimate
		$k_\infty = 6.84 \times 10^{-18}\,T^{1.646}\,e^{(-1258/T)}$	Westmoreland *et al.* (1989)
R520	$C_4H_5 + C_2H_2 \longrightarrow C_6H_6 + H$	$5.31 \times 10^{-8}\,T^{-1.11}\,e^{(-4568/T)}$	Westmoreland *et al.* (1989)
R522	$C_6H_3 + H_2 \longrightarrow C_6H_4 + H$	$1.0 \times 10^{-21}\,T^{2}\,e^{(-3500/T)}$	Estimate
R524	$C_6H_5 + H_2 \longrightarrow C_6H_6 + H$	$9.48 \times 10^{-20}\,T^{2.43}\,e^{(-3159/T)}$	Mebel *et al.* (1997)
R526	$C_6H_5 + CH_4 \longrightarrow C_6H_6 + CH_3$	$3.32 \times 10^{-12}\,e^{(-4329/T)}$	Heckmann *et al.* (1996)
R528	$l{-}C_6H_6 \longrightarrow C_6H_6$	1.0×10^{3}	Estimate
R530	$l{-}C_6H_6 + H \longrightarrow C_6H_6 + H$	$1.44 \times 10^{-7}\,T^{-1.34}\,e^{(-1761/T)}$	Wang and Frenklach (1997)
R532	$O + H \xrightarrow{M} OH$	$k_0 = 1.3 \times 10^{-29}\,T^{-1}$	Tsang and Hampson (1986)
		$k_\infty = 1.0 \times 10^{-11}$	Estimate
R534	$O + H_2 \longrightarrow OH + H$	$8.49 \times 10^{-20}\,T^{2.67}\,e^{(-3160/T)}$	Baulch *et al.* (1992)
R536	$O + CH \longrightarrow CO + H$	6.6×10^{-11}	Baulch *et al.* (1992)
R538	$O + {}^3CH_2 \longrightarrow CO + 2\,H$	1.2×10^{-10}	Baulch *et al.* (1992)
R540	$O + {}^3CH_2 \longrightarrow CO + H_2$	8.0×10^{-11}	Baulch *et al.* (1992)
R542	$O + CH_3 \longrightarrow H_2CO + H$	1.4×10^{-10}	Atkinson *et al.* (1992)
R544	$O + CH_4 \longrightarrow OH + CH_3$	$1.5 \times 10^{-15}\,T^{1.56}\,e^{(-4270/T)}$	Baulch *et al.* (1992)
R546	$O + C_2H \longrightarrow CO + CH$	1.7×10^{-11}	Baulch *et al.* (1992)
R548	$O + C_2H_2 \longrightarrow CO + {}^3CH_2$	$2.04 \times 10^{-18}\,T^{2.1}\,e^{(-786/T)}$	Based on Baulch *et al.* (1994) and
R550	$O + C_2H_2 \longrightarrow HCCO + H$	$9.96 \times 10^{-18}\,T^{2.1}\,e^{(-786/T)}$	Peeters *et al.* (1994)
R552	$O + C_2H_3 \longrightarrow H_2CCO + H$	1.25×10^{-11}	Based on Baulch *et al.* (1994)
R554	$O + C_2H_3 \longrightarrow OH + C_2H_2$	1.25×10^{-11}	Based on Baulch *et al.* (1994)
R556	$O + C_2H_3 \longrightarrow CO + CH_3$	1.25×10^{-11}	Based on Baulch *et al.* (1994)
R558	$O + C_2H_3 \longrightarrow HCO + {}^3CH_2$	1.25×10^{-11}	Based on Baulch *et al.* (1994)
R560	$O + C_2H_4 \longrightarrow HCO + CH_3$	$1.31 \times 10^{-17}\,T^{1.88}\,e^{(-92/T)}$	Based on Baulch *et al.* (1994)
R562	$O + C_2H_4 \longrightarrow C_2H_3O + H$	$7.2 \times 10^{-18}\,T^{1.88}\,e^{(-92/T)}$	Based on Baulch *et al.* (1994)
R564	$O + C_2H_4 \longrightarrow H_2CCO + H_2$	$1.12 \times 10^{-18}\,T^{1.88}\,e^{(-92/T)}$	Based on Baulch *et al.* (1994)
R566	$O + C_2H_4 \longrightarrow H_2CO + {}^3CH_2$	$1.12 \times 10^{-18}\,T^{1.88}\,e^{(-92/T)}$	Based on Baulch *et al.* (1994)
R568	$O + C_2H_5 \longrightarrow CH_3CHO + H$	8.3×10^{-11}	Baulch *et al.* (1992)
R570	$O + C_2H_5 \longrightarrow H_2CO + CH_3$	1.7×10^{-11}	Baulch *et al.* (1992)
R572	$O + C_2H_6 \longrightarrow OH + C_2H_5$	$5.12 \times 10^{-22}\,T^{3.6}\,e^{(-1882/T)}$	literature review
R574	$O + C_6H_6 \longrightarrow OH + C_6H_5$	$3.32 \times 10^{-11}\,e^{(-7400/T)}$	Leidreiter and Wagner (1989)
R576	$2\,O \xrightarrow{M} O_2$	$k_0 = 5.2 \times 10^{-35}\,e^{(900/T)}$	Tsang and Hampson (1986)
R578	$O + OH \longrightarrow O_2 + H$	$2.0 \times 10^{-11}\,e^{(112/T)}$	Baulch *et al.* (1994)
R580	$O + CO \xrightarrow{M} CO_2$	$k_0 = 1.7 \times 10^{-33}\,e^{(-1510/T)}$	Tsang and Hampson (1986)
		$k_\infty = 2.66 \times 10^{-14}\,e^{(-1459/T)}$	Simonaitis and Heicklen (1972)
R582	$O + CO_2 \longrightarrow CO + O_2$	$2.8 \times 10^{-11}\,e^{(-26500/T)}$	Tsang and Hampson (1986)

	Reaction	Rate constant	Reference
R584	$O + HCO \longrightarrow OH + CO$	5.0×10^{-11}	Baulch *et al.* (1992)
R586	$O + HCO \longrightarrow CO_2 + H$	5.0×10^{-11}	Baulch *et al.* (1992)
R588	$O + H_2CO \longrightarrow OH + HCO$	$6.85 \times 10^{-13}\, T^{0.57}\, e^{(-1390/T)}$	Baulch *et al.* (1992)
R590	$O + CH_2OH \longrightarrow OH + H_2CO$	1.5×10^{-10}	Grotheer *et al.* (1989)
R592	$O + CH_3O \longrightarrow O_2 + CH_3$	$3.55 \times 10^{-11}\, e^{(-239/T)}$	Cobos and Troe (1985)
R594	$O + CH_3O \longrightarrow OH + H_2CO$	3.0×10^{-12}	Baulch *et al.* (1992)
R596	$O + CH_3OH \longrightarrow OH + CH_2OH$	$1.63 \times 10^{-11}\, e^{(-2267/T)}$	Failes *et al.* (1982)
R598	$O + HCCO \longrightarrow 2\,CO + H$	1.6×10^{-10}	Baulch *et al.* (1992)
R600	$O + H_2CCO \longrightarrow CO + H_2CO$	$1.3 \times 10^{-12}\, e^{(-680/T)}$	Baulch *et al.* (1992)
R602	$O + H_2CCO \longrightarrow 2\,HCO$	$1.3 \times 10^{-12}\, e^{(-680/T)}$	Baulch *et al.* (1992)
R604	$O + H_2CCO \longrightarrow CO + HCO + H$	$1.3 \times 10^{-12}\, e^{(-680/T)}$	Baulch *et al.* (1992)
R606	$O + C_2H_3O \longrightarrow CO_2 + CH_3$	2.4×10^{-10}	Miyoshi *et al.* (1989)
R608	$O + C_2H_3O \longrightarrow OH + H_2CCO$	7.0×10^{-11}	Miyoshi *et al.* (1989)
R610	$O + CH_3CHO \longrightarrow OH + C_2H_3O$	$9.7 \times 10^{-12}\, e^{(-910/T)}$	Baulch *et al.* (1992)
R612	$O + C_2H_4OH \longrightarrow OH + CH_3CHO$	1.5×10^{-10}	Herron (1988)
R614	$O + HO_2 \longrightarrow O_2 + OH$	$2.7 \times 10^{-11}\, e^{(224/T)}$	Atkinson *et al.* (1992)
R616	$O + H_2O_2 \longrightarrow OH + HO_2$	$1.1 \times 10^{-12}\, e^{(-2000/T)}$	Baulch *et al.* (1992)
R618	$O(^1D) + H_2 \longrightarrow OH + H$	1.1×10^{-10}	Atkinson *et al.* (1992)
R620	$O(^1D) + CH_4 \longrightarrow OH + CH_3$	1.35×10^{-10}	Atkinson *et al.* (1992)
R622	$O(^1D) + CH_4 \longrightarrow H_2CO + H_2$	1.5×10^{-11}	Atkinson *et al.* (1992)
R624	$O(^1D) + C_2H_6 \longrightarrow OH + C_2H_5$	6.3×10^{-10}	Matsumi *et al.* (1993)
R626	$O(^1D) + H_2O \longrightarrow 2\,OH$	2.2×10^{-10}	DeMore *et al.* (1987)
R628	$O(^1D) + CO_2 \longrightarrow O + CO_2$	$7.4 \times 10^{-11}\, e^{(120/T)}$	Atkinson *et al.* (1992)
R630	$O_2 + H \xrightarrow{M} HO_2$	$k_0 = 5.8 \times 10^{-30}\, T^{-0.8}$	Baulch *et al.* (1994)
		$k_\infty = 1.46 \times 10^{-13}\, T^{1}\, e^{(-224/T)}$	Duchovic *et al.* (1996)
R632	$O_2 + C \longrightarrow O + CO$	1.6×10^{-11}	Dorthe *et al.* (1991)
R634	$O_2 + CH \longrightarrow OH + CO$	2.75×10^{-11}	Baulch *et al.* (1992)
R636	$O_2 + CH \longrightarrow O + HCO$	2.75×10^{-11}	Baulch *et al.* (1992)
R638	$O_2 + {}^3CH_2 \longrightarrow OH + CO + H$	1.0×10^{-12}	Tsang and Hampson (1986)
R640	$O_2 + {}^3CH_2 \longrightarrow H_2O + CO$	4.0×10^{-13}	Tsang and Hampson (1986)
R642	$O_2 + {}^1CH_2 \longrightarrow H + OH + CO$	4.6×10^{-11}	Wang and Frenklach (1997)
R644	$O_2 + {}^1CH_2 \longrightarrow H_2O + CO$	2.0×10^{-11}	Wang and Frenklach (1997)
R646	$O_2 + CH_3 \longrightarrow H_2CO + OH$	$5.64 \times 10^{-13}\, e^{(-4500/T)}$	Zellner and Ewig (1988)
R648	$O_2 + CH_4 \longrightarrow HO_2 + CH_3$	$6.6 \times 10^{-11}\, e^{(-28630/T)}$	Baulch *et al.* (1992)
R650	$O_2 + C_2H \longrightarrow CO + HCO$	$6.25 \times 10^{-11}\, T^{-0.16}$	Based on Thiesemann and Taatjes (1997)
R652	$O_2 + C_2H \longrightarrow HCCO + O$	$1.56 \times 10^{-11}\, T^{-0.16}$	Based on Thiesemann and Taatjes (1997)
R654	$O_2 + C_2H_3 \longrightarrow H_2CO + HCO$	$7.6 \times 10^{-8}\, T^{-1.39}\, e^{(-510/T)}$	Mebel *et al.* (1996a)
R656	$O_2 + C_2H_6 \longrightarrow HO_2 + C_2H_5$	$1.0 \times 10^{-10}\, e^{(-26100/T)}$	Baulch *et al.* (1992)
R658	$O_2 + HCO \longrightarrow CO + HO_2$	5.5×10^{-12}	Atkinson *et al.* (1997)
R660	$OH + H \xrightarrow{M} H_2O$	$k_0 = 6.1 \times 10^{-26}\, T^{-2}$	Baulch *et al.* (1992)
		$k_\infty = 2.69 \times 10^{-10}\, e^{(-75/T)}$	Cobos and Troe (1985)
R662	$OH + H_2 \longrightarrow H_2O + H$	$1.57 \times 10^{-11}\, e^{(-2346/T)}$	$T < 600$ K, Fisher and Michael (1990)
		$1.75 \times 10^{-16}\, T^{1.63}\, e^{(-1753/T)}$	for $T \geq 600$ K
R664	$OH + {}^3CH_2 \longrightarrow H_2CO + H$	3.0×10^{-11}	Tsang and Hampson (1986)

TABLE S1 (*cont.*)

Reaction				Rate constant	Reference
R666	$OH + CH_3$	$\longrightarrow$	$H_2O + {}^3CH_2$	$7.13 \times 10^{-20}\, T^{2.568}\, e^{(-2012/T)}$	Jasper *et al.* (2007)
R668	$OH + CH_3$	$\longrightarrow$	$H_2O + {}^1CH_2$	$k_0 = 3.714 \times 10^{-8}\, T^{-1.172}\, e^{(-132/T)}$	
				$+ 1.535 \times 10^{-19} T^{2.127}\, e^{(-547.8/T)}$	Jasper *et al.* (2007)
				$k_\infty = 7.976 \times 10^{-4}\, T^{4.096}\, e^{(625/T)}$	Jasper *et al.* (2007)
				$F_c = 0.4863 e^{(-T/321.4)}$	
				$+ 0.5137 e^{(-T/30000)}$	
				$+ e^{(-2804/T)}$	Jasper *et al.* (2007)
				$N = 0.75 - 1.27 \log F_c$	
				$\log F = \dfrac{\log F_c}{1 + [\log(k_0[M]/k_\infty)/N]^2}$	
				$k = k_0 k_\infty F/(k_\infty + k_0[M])$	
R670	$OH + CH_3$	$\longrightarrow$	$CH_2OH + H$	$k_0 = 1.092 \times 10^{-14}\, T^{0.996}\, e^{(-1606/T)}$	Jasper *et al.* (2007)
				$k_\infty = 5.864 \times 10^{-6}\, T^{5.009}\, e^{(-949.4/T)}$	Jasper *et al.* (2007)
				$F_c = 0.8622 e^{(-T/9321)}$	
				$+ 0.1378 e^{(-T/361.8)}$	
				$+ e^{(-3125/T)}$	Jasper *et al.* (2007)
				$N = 0.75 - 1.27 \log F_c$	
				$\log F = \dfrac{\log F_c}{1 + [\log(k_0[M]/k_\infty)/N]^2}$	
				$k = k_0 k_\infty F/(k_\infty + k_0[M])$	
R672	$OH + CH_3$	$\longrightarrow$	$H_2CO + H_2$	$k_0 = 2.359 \times 10^{-10}\, T^{-1.234}\, e^{(19.57/T)}$	
				$+ 3.141 \times 10^{3} T^{-4.484}\, e^{(-9188/T)}$	Jasper *et al.* (2007)
				$k_\infty = 5.876 \times 10^{-14}\, T^{6.721}\, e^{(1521/T)}$	Jasper *et al.* (2007)
				$F_c = 0.5 e^{(-T/22{,}122)}$	
				$+ 0.5 e^{(-T/174.9)}$	
				$+ e^{(-3047/T)}$	Jasper *et al.* (2007)
				$N = 0.75 - 1.27 \log F_c$	
				$\log F = \dfrac{\log F_c}{1 + [\log(k_0[M]/k_\infty)/N]^2}$	
				$k = k_0 k_\infty F/(k_\infty + k_0[M])$	
R674	$OH + CH_3$	$\xrightarrow{M}$	CH_3OH	$k_0 = 1.932 \times 10^{3}\, T^{-9.88}\, e^{(-7544/T)}$	
				$+ 5.109 \times 10^{-11} T^{-6.25}\, e^{(-1433/T)}$	Jasper *et al.* (2007)
				$k_\infty = 1.031 \times 10^{-10}\, T^{-0.018}\, e^{(16.74/T)}$	Jasper *et al.* (2007)
				$F_c = 0.1855 e^{(-T/155.8)}$	
				$+ 0.8145 e^{(-T/1675)}$	
				$+ e^{(-4531/T)}$	Jasper *et al.* (2007)
				$N = 0.75 - 1.27 \log F_c$	
				$\log F = \dfrac{\log F_c}{1 + [\log(k_0[M]/k_\infty)/N]^2}$	
				$k = k_0 k_\infty F/(k_\infty + k_0[M])$	
R676	$OH + CH_4$	$\longrightarrow$	$H_2O + CH_3$	$1.85 \times 10^{-20}\, T^{2.82}\, e^{(-987/T)}$	Gierczak *et al.* (1997)
R678	$OH + C_2H$	$\longrightarrow$	$HCCO + H$	3.0×10^{-11}	Estimate
R680	$OH + C_2H_2$	$\longrightarrow$	$CO + CH_3$	$8.04 \times 10^{-28}\, T^{4}\, e^{(1006/T)}$	Miller and Melius (1989)
R682	$OH + C_2H_2$	$\longrightarrow$	$H_2CCO + H$	$3.64 \times 10^{-28}\, T^{4.5}\, e^{(503/T)}$	Miller and Melius (1989)
R684	$OH + C_2H_2$	$\longrightarrow$	$H_2O + C_2H$	$5.62 \times 10^{-17}\, T^{2}\, e^{(-7045/T)}$	Miller and Melius (1989)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R686	$OH + C_2H_2 \xrightarrow{M} C_2H_3O$	$k_0 = 2.05 \times 10^{-21}\, T^{-3.1}\, e^{(-910/T)}$	Atkinson *et al.* (1997)
		$k_\infty = 3.8 \times 10^{-11}\, e^{(-910/T)}$	Fulle *et al.* (1997)
		$F_c = 0.17e^{(-51/T)} + e^{(-T/204)}$	Fulle *et al.* (1997)
R688	$OH + C_2H_3 \longrightarrow H_2O + C_2H_2$	5.0×10^{-11}	Tsang and Hampson (1986)
R690	$OH + C_2H_3 \xrightarrow{M} CH_3CHO$	$k_0 = 1.0 \times 10^{-25}\, T^{-2}$	Estimate
		$k_\infty = 5.0 \times 10^{-11}$	Tsang and Hampson (1986)
R692	$OH + C_2H_4 \longrightarrow H_2O + C_2H_3$	$3.4 \times 10^{-11}\, e^{(-2990/T)}$	Baulch *et al.* (1992)
R694	$OH + C_2H_4 \xrightarrow{M} C_2H_4OH$	$k_0 = 1.08 \times 10^{-20}\, T^{-3.4}$	Fulle *et al.* (1997)
		$k_\infty = 1.0 \times 10^{-11}$	Fulle *et al.* (1997)
		$F_c = 0.21e^{(-220/T)} + e^{(-T/305)}$	Fulle *et al.* (1997)
R696	$OH + C_2H_5 \longrightarrow H_2O + C_2H_4$	4.0×10^{-11}	Tsang and Hampson (1986)
R698	$OH + C_2H_6 \longrightarrow H_2O + C_2H_5$	$1.2 \times 10^{-17}\, T^{2}\, e^{(-435/T)}$	Baulch *et al.* (1994)
R700	$OH + CH_3C_2H \longrightarrow H_2O + C_3H_3$	$7.1 \times 10^{-13}\, e^{(424/T)}$	Boodaghians *et al.* (1987)
R702	$OH + C_3H_5 \longrightarrow H_2O + CH_3C_2H$	1.0×10^{-11}	Tsang (1991)
R704	$OH + C_3H_6 \longrightarrow H_2O + C_3H_5$	$5.19 \times 10^{-18}\, T^{2}\, e^{(150/T)}$	Tsang (1991)
R706	$OH + C_3H_8 \longrightarrow H_2O + C_3H_7$	$1.04 \times 10^{-16}\, T^{1.72}\, e^{(-145/T)}$	Droege and Tully (1986)
R708	$OH + C_6H_6 \longrightarrow H_2O + C_6H_5$	$2.7 \times 10^{-16}\, T^{1.42}\, e^{(-730/T)}$	Baulch *et al.* (1994)
R710	$2\,OH \longrightarrow O + H_2O$	$2.5 \times 10^{-15}\, T^{1.14}\, e^{(-50/T)}$	Baulch *et al.* (1994)
R712	$2\,OH \xrightarrow{M} H_2O_2$	$k_0 = 6.1 \times 10^{-29}\, T^{-0.76}$	Baulch *et al.* (1994)
		$k_\infty = 1.2 \times 10^{-10}\, T^{-0.37}$	Baulch *et al.* (1994)
R714	$OH + CO \longrightarrow CO_2 + H$	$5.49 \times 10^{-18}\, T^{1.55}\, e^{(402/T)}$	Lissianski *et al.* (1995)
		1.449×10^{-13} for $T \leq 300$ K	Est., literature review
R716	$OH + HCO \longrightarrow H_2O + CO$	1.7×10^{-10}	Baulch *et al.* (1992)
R718	$OH + H_2CO \longrightarrow H_2O + HCO$	$5.7 \times 10^{-15}\, T^{1.18}\, e^{(225/T)}$	Baulch *et al.* (1994)
R720	$OH + CH_2OH \longrightarrow H_2O + H_2CO$	4.0×10^{-11}	Tsang (1987)
R722	$OH + CH_3O \longrightarrow H_2O + H_2CO$	3.0×10^{-11}	Tsang and Hampson (1986)
R724	$OH + CH_3OH \longrightarrow H_2O + CH_2OH$	$2.39 \times 10^{-18}\, T^{2}\, e^{(423/T)}$	Li and Williams (1996)
R726	$OH + CH_3OH \longrightarrow H_2O + CH_3O$	$2.39 \times 10^{-19}\, T^{2}\, e^{(423/T)}$	Based on Dóbé *et al.* (1994a)
R728	$OH + H_2CCO \longrightarrow CO + CH_2OH$	1.0×10^{-11}	Grussdorf *et al.* (1994)
R730	$OH + H_2CCO \longrightarrow HCO + H_2CO$	3.0×10^{-13}	Grussdorf *et al.* (1994)
R732	$OH + C_2H_3O \longrightarrow H_2O + H_2CCO$	2.0×10^{-11}	Tsang and Hampson (1986)
R734	$OH + CH_3CHO \longrightarrow H_2O + C_2H_3O$	$2.58 \times 10^{-18}\, T^{2.2}\, e^{(-503/T)}$	Taylor *et al.* (1996)
R736	$OH + HO_2 \longrightarrow H_2O + O_2$	$4.8 \times 10^{-11}\, e^{(250/T)}$	Baulch *et al.* (1994)
R738	$OH + H_2O_2 \longrightarrow H_2O + HO_2$	$1.3 \times 10^{-11}\, e^{(-670/T)}$	Baulch *et al.* (1994)
R740	$H_2O + CH \longrightarrow H_2CO + H$	$9.5 \times 10^{-12}\, e^{(380/T)}$	Zabarnick *et al.* (1988)
R742	$H_2O + {}^1CH_2 \longrightarrow {}^3CH_2 + H_2O$	5.0×10^{-11}	Wang and Frenklack (1997)
R744	$CO \xrightarrow{M} C + O$	$7.13 \times 10^{3}\, T^{-3.1}\, e^{(-129122/T)}$	Mick *et al.* (1993)
R746	$CO + H \xrightarrow{M} HCO$	$k_0 = 5.3 \times 10^{-34}\, e^{(-370/T)}$	Baulch *et al.* (1994)
		$k_\infty = 1.96 \times 10^{-13}\, e^{(-1366/T)}$	Arai *et al.* (1981)
R748	$CO + H_2 \xrightarrow{M} H_2CO$	$k_0 = 2.796 \times 10^{-20}\, T^{-3.42}\, e^{(-42445/T)}$	Wang and Frenklach (1997)
		$k_\infty = 7.14 \times 10^{-17}\, T^{1.5}\, e^{(-40055/T)}$	Wang and Frenklach (1997)
R750	$CO + CH \xrightarrow{M} HCCO$	$k_0 = 6.28 \times 10^{-24}\, T^{-2.5}$	Brownsword *et al.* (1996b)
		$k_\infty = 6.0 \times 10^{-6}\, T^{-2.4}$	Taatjes (1997)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R752	$CO + {}^1CH_2 \longrightarrow {}^3CH_2 + CO$	5.0×10^{-11}	Wang and Frenklach (1997)
R754	$CO + {}^3CH_2 \xrightarrow{M} H_2CCO$	$k_0 = 7.4 \times 10^{-15}\, T^{-5.11}\, e^{(-3570/T)}$	Wang and Frenklach (1997)
		$k_\infty = 1.35 \times 10^{-12}\, T^{0.5}\, e^{(-2269/T)}$	Wang and Frenklach (1997)
R756	$CO + CH_3 \xrightarrow{M} C_2H_3O$	$k_0 = 3.0 \times 10^{-34}\, e^{(-1910/T)}$	Baulch *et al.* (1994)
		$k_\infty = 8.4 \times 10^{-13}\, e^{(-3460/T)}$	Baulch *et al.* (1994)
R758	$CO + HO_2 \longrightarrow CO_2 + OH$	$3.45 \times 10^{-12}\, T^{0.5}\, e^{(-11500/T)}$	Volman (1996)
R760	$CO_2 + CH \longrightarrow CO + HCO$	$5.7 \times 10^{-12}\, e^{(-345/T)}$	Baulch *et al.* (1994)
R762	$CO_2 + {}^3CH_2 \longrightarrow CO + H_2CO$	5.0×10^{-15}	Based on Darwin and Moore (1995)
R764	$HCO + H \longrightarrow CO + H_2$	1.5×10^{-10}	Baulch *et al.* (1994)
R766	$HCO + H \xrightarrow{M} H_2CO$	$k_0 = 2.475 \times 10^{-26}\, T^{-2}$	literature review
R768	$HCO + {}^3CH_2 \longrightarrow CO + CH_3$	3.0×10^{-11}	Tsang and Hampson (1986)
R770	$HCO + CH_3 \longrightarrow CO + CH_4$	4.4×10^{-11}	Mulenko (1987)
R772	$HCO + CH_3 \xrightarrow{M} CH_3CHO$	$k_0 = 1.0 \times 10^{-28}\, e^{(1200/T)}$	Estimate
		$k_\infty = 5.0 \times 10^{-11}$	Callear and Cooper (1990)
R774	$HCO + C_2H \longrightarrow CO + C_2H_2$	1.0×10^{-10}	Tsang and Hampson (1986)
R776	$HCO + C_2H_3 \longrightarrow CO + C_2H_4$	1.5×10^{-10}	Tsang and Hampson (1986)
R778	$HCO + C_2H_5 \longrightarrow CO + C_2H_6$	7.2×10^{-11}	Baggott *et al.* (1987)
R780	$2\,HCO \longrightarrow CO + H_2CO$	5.0×10^{-11}	Baulch *et al.* (1992)
R782	$HCO + CH_2OH \longrightarrow 2\,H_2CO$	3.0×10^{-10}	Tsang (1987)
R784	$HCO + CH_2OH \longrightarrow CO + CH_3OH$	2.0×10^{-10}	Tsang (1987)
R786	$HCO + CH_3O \longrightarrow CO + CH_3OH$	1.5×10^{-10}	Tsang and Hampson (1986)
R788	$HCO + C_2H_3O \longrightarrow CO + CH_3CHO$	1.5×10^{-11}	Tsang and Hampson (1986)
R790	$H_2CO + H \longrightarrow HCO + H_2$	$9.53 \times 10^{-17}\, T^{1.9}\, e^{(-1379/T)}$	Irdam *et al.* (1993)
R792	$H_2CO + CH \longrightarrow H_2CCO + H$	$1.57 \times 10^{-10}\, e^{(260/T)}$	Based on Zabarnick *et al.* (1988)
R794	$H_2CO + CH_3 \longrightarrow HCO + CH_4$	$1.3 \times 10^{-31}\, T^{6.1}\, e^{(-990/T)}$	Baulch *et al.* (1994)
R796	$H_2CO + C_2H_3 \longrightarrow HCO + C_2H_4$	$9 \times 10^{-21}\, T^{2.81}\, e^{(-2950/T)}$	Tsang and Hampson (1986)
R798	$H_2CO + C_2H_5 \longrightarrow HCO + C_2H_6$	$9.13 \times 10^{-21}\, T^{2.81}\, e^{(-2950/T)}$	Tsang and Hampson (1986)
R800	$H_2CO + CH_2OH \longrightarrow HCO + CH_3OH$	$9.13 \times 10^{-21}\, T^{2.8}\, e^{(-2950/T)}$	Tsang (1987)
R802	$H_2CO + CH_3O \longrightarrow HCO + CH_3OH$	$1.7 \times 10^{-13}\, e^{(-1500/T)}$	Tsang and Hampson (1986)
R804	$CH_2OH \xrightarrow{M} H + H_2CO$	$k_0 = 1.66 \times 10^{-10}\, e^{(-12630/T)}$	Cribb *et al.* (1992)
R806	$CH_2OH + H \longrightarrow H_2CO + H_2$	1.66×10^{-11}	Cribb *et al.* (1992)
R808	$CH_2OH + {}^3CH_2 \longrightarrow OH + C_2H_4$	4.0×10^{-11}	Tsang (1987)
R810	$CH_2OH + {}^3CH_2 \longrightarrow H_2CO + CH_3$	2.0×10^{-12}	Tsang (1987)
R812	$CH_2OH + CH_3 \longrightarrow H_2CO + CH_4$	1.4×10^{-10}	Pagsberg *et al.* (1988)
R814	$CH_2OH + C_2H \longrightarrow OH + C_3H_3$	2.0×10^{-11}	Tsang (1987)
R816	$CH_2OH + C_2H \longrightarrow H_2CO + C_2H_2$	6.0×10^{-11}	Tsang (1987)
R818	$CH_2OH + C_2H_2 \longrightarrow H_2CO + C_2H_3$	$1.2 \times 10^{-12}\, e^{(-4531/T)}$	Tsang (1987)
R820	$CH_2OH + C_2H_3 \longrightarrow OH + C_3H_5$	2.0×10^{-11}	Tsang (1987)
R822	$CH_2OH + C_2H_3 \longrightarrow H_2CO + C_2H_4$	5.0×10^{-11}	Tsang (1987)
R824	$CH_2OH + C_2H_5 \longrightarrow CH_3OH + C_2H_4$	4.0×10^{-12}	Tsang (1987)
R826	$CH_2OH + C_2H_5 \longrightarrow H_2CO + C_2H_6$	4.0×10^{-12}	Tsang (1987)
R828	$2\,CH_2OH \longrightarrow H_2CO + CH_3OH$	8.0×10^{-12}	Tsang (1987)
R830	$CH_2OH + CH_3O \longrightarrow H_2CO + CH_3OH$	4.0×10^{-11}	Tsang (1987)

	Reaction	Rate constant	Reference
R832	$CH_3O \xrightarrow{M} H + H_2CO$	$k_0 = 1.4 \times 10^{-6}\, T^{-1.2}\, e^{(-7800/T)}$	Page *et al.* (1989)
		$k_\infty = 1.56 \times 10^{15}\, T^{-0.39}\, e^{(-13285/T)}$	Curran (2006)
R834	$CH_3O + H \longrightarrow OH + CH_3$	$7.52 \times 10^{-11}\, e^{(-375/T)}$	Dóbé *et al.* (1991)
R836	$CH_3O + H \longrightarrow H_2CO + H_2$	$3.38 \times 10^{-11}\, e^{(-375/T)}$	Dóbé *et al.* (1991)
R838	$CH_3O + {}^3CH_2 \longrightarrow H_2CO + CH_3$	3.0×10^{-11}	Tsang (1987)
R840	$CH_3O + CH_3 \longrightarrow H_2CO + CH_4$	4.0×10^{-11}	Tsang and Hampson (1986)
R842	$CH_3O + C_2H \longrightarrow H_2CO + C_2H_2$	4.0×10^{-11}	Tsang and Hampson (1986)
R844	$CH_3O + C_2H_3 \longrightarrow H_2CO + C_2H_4$	4.0×10^{-11}	Tsang and Hampson (1986)
R846	$CH_3O + C_2H_5 \longrightarrow H_2CO + C_2H_6$	4.0×10^{-11}	Tsang and Hampson (1986)
R848	$2\,CH_3O \longrightarrow H_2CO + CH_3OH$	1.0×10^{-10}	Tsang and Hampson (1986)
R850	$CH_3O + CH_3OH \longrightarrow CH_2OH + CH_3OH$	$5.0 \times 10^{-13}\, e^{(-2050/T)}$	Tsang (1987)
R852	$CH_3O + C_2H_3O \longrightarrow H_2CCO + CH_3OH$	1.0×10^{-11}	Tsang and Hampson (1986)
R854	$CH_3O + C_2H_3O \longrightarrow H_2CO + CH_3CHO$	1.0×10^{-11}	Tsang and Hampson (1986)
R856	$CH_3O + CH_3CHO \longrightarrow C_2H_3O + CH_3OH$	6.0×10^{-15}	Estimate
R858	$CH_3OH \xrightarrow{M} H_2O + {}^1CH_2$	$k_0 = 1.484 \times 10^{30}\, T^{-10.2}\, e^{(-52454/T)}$	
		$+ 1.223 \times 10^{17} T^{-6.577}\, e^{(-48007/T)}$	Jasper *et al.* (2007)
		$k_\infty = 3.121 \times 10^{18}\, T^{-1.017}\, e^{(-46156/T)}$	Jasper *et al.* (2007)
		$F_c = 0.9922 e^{(-T/943)}$	
		$+ 7.8 \times 10^{-3} e^{(-T/47310)}$	
		$+ e^{(-47110/T)}$	Jasper *et al.* (2007)
		$N = 0.75 - 1.27 \log F_c$	
		$\log F = \dfrac{\log F_c}{1 + [\log(k_0[M]/k_\infty)/N]^2}$	
		$k = k_0 k_\infty F/(k_\infty + k_0[M])$	
R860	$CH_3OH + H \longrightarrow H_2O + CH_3$	$4.91 \times 10^{-19} T^{2.485}\, e^{(-10380/T)}$	this work
R862	$CH_3OH + H \longrightarrow CH_3O + H_2$	$6.82 \times 10^{-20} T^{2.658}\, e^{(-4643/T)}$	this work
R864	$CH_3OH + H \longrightarrow CH_2OH + H_2$	$1.09 \times 10^{-19} T^{2.728}\, e^{(-2240/T)}$	this work
R866	$CH_3OH + {}^3CH_2 \longrightarrow CH_2OH + CH_3$	$5.29 \times 10^{-23} T^{3.2}\, e^{(-3609/T)}$	Tsang (1987)
R868	$CH_3OH + {}^3CH_2 \longrightarrow CH_3O + CH_3$	$2.39 \times 10^{-23} T^{3.1}\, e^{(-3490/T)}$	Tsang (1987)
R870	$CH_3OH + CH_3 \longrightarrow CH_2OH + CH_4$	$5.29 \times 10^{-23} T^{3.2}\, e^{(-3609/T)}$	Tsang (1987)
R872	$CH_3OH + CH_3 \longrightarrow CH_3O + CH_4$	$2.39 \times 10^{-23} T^{3.1}\, e^{(-3490/T)}$	Tsang (1987)
R874	$CH_3OH + C_2H \longrightarrow CH_2OH + C_2H_2$	1.0×10^{-11}	Tsang (1987)
R876	$CH_3OH + C_2H \longrightarrow CH_3O + C_2H_2$	2.0×10^{-12}	Tsang (1987)
R878	$CH_3OH + C_2H_3 \longrightarrow CH_2OH + C_2H_4$	$5.29 \times 10^{-23} T^{3.2}\, e^{(-3609/T)}$	Tsang (1987)
R880	$CH_3OH + C_2H_3 \longrightarrow CH_3O + C_2H_4$	$2.39 \times 10^{-23} T^{3.1}\, e^{(-3490/T)}$	Tsang (1987)
R882	$CH_3OH + C_2H_5 \longrightarrow CH_2OH + C_2H_6$	$5.29 \times 10^{-23} T^{3.2}\, e^{(-4610/T)}$	Tsang (1987)
R884	$CH_3OH + C_2H_5 \longrightarrow CH_3O + C_2H_6$	$2.39 \times 10^{-23} T^{3.1}\, e^{(-4500/T)}$	Tsang (1987)
R886	$HCCO + H \longrightarrow CO + {}^3CH_2$	2.5×10^{-10}	Frank *et al.* (1988)
R888	$HCCO + H_2 \longrightarrow H_2CCO + H$	$3.0 \times 10^{-14} T^{0.7}\, e^{(-6600/T)}$	Estimate
R890	$H_2CCO + H \longrightarrow CO + CH_3$	$3.0 \times 10^{-11}\, e^{(-1700/T)}$	Baulch *et al.* (1992)
R892	$C_2H_3O + H \longrightarrow HCO + CH_3$	3.57×10^{-11}	Bartels *et al.* (1991) and
R894	$C_2H_3O + H \longrightarrow H_2CCO + H_2$	1.92×10^{-11}	Ohmori *et al.* (1990)
R896	$C_2H_3O + {}^3CH_2 \longrightarrow H_2CCO + CH_3$	3.0×10^{-11}	Tsang and Hampson (1986)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R898	$C_2H_3O + CH_3 \longrightarrow CO + C_2H_6$	4.9×10^{-11}	Adachi *et al.* (1981) and
R900	$C_2H_3O + CH_3 \longrightarrow H_2CCO + CH_4$	1.0×10^{-11}	Hassinen *et al.* (1990)
R902	$C_2H_3O + C_2H \longrightarrow H_2CCO + C_2H_2$	3.0×10^{-11}	Tsang and Hampson (1986)
R904	$2\,C_2H_3O \longrightarrow H_2CCO + CH_3CHO$	1.49×10^{-11}	Hassinen *et al.* (1990)
R906	$CH_3CHO + H \longrightarrow C_2H_3O + H_2$	$2.23 \times 10^{-11}\,e^{(-1661/T)}$	Whytock *et al.* (1976a)
R908	$CH_3CHO + {}^3CH_2 \longrightarrow C_2H_3O + CH_3$	$2.76 \times 10^{-12}\,e^{(-1768/T)}$	Böhland *et al.* (1985a)
R910	$CH_3CHO + CH_3 \longrightarrow C_2H_3O + CH_4$	$3.3 \times 10^{-30}\,T^{5.64}\,e^{(-1240/T)}$	Baulch *et al.* (1992)
R912	$CH_3CHO + C_2H_3 \longrightarrow C_2H_3O + C_2H_4$	$1.35 \times 10^{-13}\,e^{(-1852/T)}$	Scherzer *et al.* (1987)
R914	$CH_3CHO + C_2H_5 \longrightarrow C_2H_3O + C_2H_6$	$2.09 \times 10^{-12}\,e^{(-4277/T)}$	Höhlein and Freeman (1970)
R916	$C_2H_4OH + H \longrightarrow CH_3CHO + H_2$	8.3×10^{-11}	Bartels *et al.* (1982)
R918	$C_2H_4OH + CH_3 \longrightarrow CH_3CHO + CH_4$	4.0×10^{-11}	Estimate
R920	$HO_2 + H \longrightarrow H_2O + O$	$5.0 \times 10^{-11}\,e^{(-866/T)}$	Baulch *et al.* (1992)
R922	$HO_2 + H \longrightarrow 2\,OH$	$2.8 \times 10^{-10}\,e^{(-440/T)}$	Baulch *et al.* (1992)
R924	$HO_2 + H \longrightarrow O_2 + H_2$	$1.11 \times 10^{-16}\,T^{1.77}\,e^{(286/T)}$	Mousavipour and Saheb (2007)
R926	$HO_2 + CH_3 \longrightarrow CH_3O + OH$	3.0×10^{-11}	Baulch *et al.* (1992)
R928	$HO_2 + CH_4 \longrightarrow H_2O_2 + CH_3$	$1.5 \times 10^{-11}\,e^{(-12440/T)}$	Baulch *et al.* (1994)
R930	$HO_2 + H_2CO \longrightarrow H_2O_2 + HCO$	$5.0 \times 10^{-12}\,e^{(-6580/T)}$	Baulch *et al.* (1994)
R932	$2\,HO_2 \longrightarrow H_2O_2 + O_2$	2.1×10^{-12}	Taatjes and Oh (1997)
R934	$H_2O_2 + H \longrightarrow H_2 + HO_2$	$2.8 \times 10^{-12}\,e^{(-1890/T)}$	Baulch *et al.* (1992)
R936	$H_2O_2 + H \longrightarrow H_2O + OH$	$1.7 \times 10^{-11}\,e^{(-1800/T)}$	Baulch *et al.* (1992)
R938	$H_2O_2 + C_2H_3 \longrightarrow HO_2 + C_2H_4$	$2.0 \times 10^{-14}\,e^{(300/T)}$	Tsang and Hampson (1986)
R940	$H_2O_2 + C_2H_5 \longrightarrow HO_2 + C_2H_6$	$1.45 \times 10^{-14}\,e^{(-490/T)}$	Tsang and Hampson (1986)
R942	$N + H_2 \longrightarrow NH + H$	$2.66 \times 10^{-10}\,e^{(-12650/T)}$	Davidson and Hanson (1990)
R944	$N + H_2 \xrightarrow{M} NH_2$	$k_0 = 8.3 \times 10^{-38}$	Est., Petrishchev and Sapozhkov (1981)
		$k_\infty = 1.94 \times 10^{-20}$	Aleksandrov *et al.* (1994)
R946	$N + CH \longrightarrow NH + C$	$7.46 \times 10^{-13}\,T^{0.65}\,e^{(-1207/T)}$	Mayer and Schieler (1966)
R948	$N + CH \longrightarrow CN + H$	$2.77 \times 10^{-10}\,T^{-0.09}$	Brownsword *et al.* (1996a)
R950	$N + {}^3CH_2 \longrightarrow HCN + H$	1.7×10^{-11}	Tsai and McFadden (1990)
R952	$N + CH_3 \longrightarrow HCN + H_2$	$4.3 \times 10^{-11}\,e^{(-420/T)}$	Marston *et al.* (1989b)
R954	$N + CH_3 \longrightarrow H_2CN + H$	$3.9 \times 10^{-10}\,e^{(-420/T)}$	Marston *et al.* (1989a)
R956	$N + CH_4 \xrightarrow{M} CH_3NH$	$k_0 = 1.0 \times 10^{-34}$	Estimate
		$k_\infty = 3.0 \times 10^{-18}$	Aleksandrov *et al.* (1989)
R958	$N + C_2 \longrightarrow CN + C$	$2.9 \times 10^{-12}\,T^{0.5}$	Est., Millar *et al.* (1991)
R960	$N + C_2H_2 \xrightarrow{M} CH_2CN$	$k_0 = 1.0 \times 10^{-30}$	Estimate
		$k_\infty = 1.0 \times 10^{-16}$	Estimate
R962	$N + C_2H_3 \longrightarrow NH + C_2H_2$	1.23×10^{-11}	Est., Payne *et al.* (1996)
R964	$N + C_2H_3 \longrightarrow CH_2CN + H$	6.16×10^{-11}	Payne *et al.* (1996)
R966	$N + C_2H_4 \longrightarrow HCN + CH_3$	$1.0 \times 10^{-14}\,e^{(-1200/T)}$	Estimate
R968	$N + C_2H_5 \longrightarrow NH + C_2H_4$	6.6×10^{-11}	Est. based on Stief *et al.* (1995) and
R970	$N + C_2H_5 \longrightarrow H_2CN + CH_3$	3.3×10^{-11}	Nesbitt *et al.* (1992)
R972	$N + C_2H_5 \longrightarrow CH_3CN + H_2$	1.1×10^{-11}	”
R974	$N + C_3H_2 \longrightarrow HC_3N + H$	$1.15 \times 10^{-12}\,T^{0.5}$	Est., Millar *et al.* (1991)
R976	$N + C_3H_3 \longrightarrow HC_3N + H_2$	$1.0 \times 10^{-13}\,T^{0.5}$	Estimate

TABLE S1 (*cont.*)

	Reaction			Rate constant	Reference
R978	$N + O$	$\xrightarrow{M}$	NO	$k_0 = 5.46 \times 10^{-33}\, e^{(155/T)}$	Campbell and Gray (1973)
R980	$N + O_2$	$\longrightarrow$	$NO + O$	$2.0 \times 10^{-18}\, T^{2.15}\, e^{(-2557/T)}$	Fernandez *et al.* (1998)
R982	$N + OH$	$\longrightarrow$	$NO + H$	$3.09 \times 10^{-9}\, T^{-0.69}\, e^{(-48/T)}$	Smith and Stewart (1994)
				4.73×10^{-11}, for $T \geq 350$ K	Est. based on Baulch *et al.* (1994)
R984	$N + CO_2$	$\longrightarrow$	$NO + CO$	$3.2 \times 10^{-13}\, e^{(-1710/T)}$	Avramenko and Krasnen'kov (1967)
R986	$N + CH_3OH$	$\longrightarrow$	$HNO + CH_3$	$4.0 \times 10^{-10}\, e^{(-4330/T)}$	Roscoe and Roscoe (1973)
R988	$N + NH$	$\longrightarrow$	$N_2 + H$	2.5×10^{-11}	Hack *et al.* (1994)
				$2.57 \times 10^{-11}\, T^{0.167}\, e^{(17.9/T)}$	Test case: fit to Caridade *et al.* (2007)
R990	$N + NH_2$	$\longrightarrow$	$N_2H + H$	1.15×10^{-10}	based on Dransfeld and Wagner (1987)
R992	$N + NH_3$	$\longrightarrow$	$N_2H + H_2$	$5.94 \times 10^{-12}\, e^{(-4830/T)}$	Est., Basevich and Vedeneev (1988)
R994	$N + NH_3$	$\longrightarrow$	N_2H_3	$1.48 \times 10^{-12}\, e^{(-4830/T)}$	Est., Basevich and Vedeneev (1988)
R996	$N + CN$	$\longrightarrow$	$N_2 + C$	1.0×10^{-10}	Whyte and Phillips (1983)
R998	$N + H_2CN$	$\longrightarrow$	$N_2 + {}^3CH_2$	$5.0 \times 10^{-12}\, e^{(-200/T)}$	Est. based on Nesbitt *et al.* (1990)
R1000	$N + H_2CN$	$\longrightarrow$	$NH + HCN$	$9.5 \times 10^{-11}\, e^{(-200/T)}$	Nesbitt *et al.* (1990)
R1002	$N + CH_2CN$	$\longrightarrow$	$H_2CN + CN$	4.4×10^{-11}	Estimate
R1004	$N + NO$	$\longrightarrow$	$N_2 + O$	$3.4 \times 10^{-11}\, e^{(-24/T)}$	Duff and Sharma (1996)
R1006	$N + NO_2$	$\longrightarrow$	$N_2O + O$	$5.8 \times 10^{-12}\, e^{(220/T)}$	Wennberg *et al.* (1994)
R1008	$N + NCO$	$\longrightarrow$	$N_2 + CO$	5.5×10^{-11}	Brownsword *et al.* (1997)
R1010	N_2	$\xrightarrow{M}$	$2\,N$	$k_0 = 1.71 \times 10^{4}\, T^{-3.33}\, e^{(-113240/T)}$	Thielen and Roth (1986)
R1012	$N_2 + H_2$	$\longrightarrow$	H_2NN	$1.776 \times 10^{-16}\, T^{1.6}\, e^{(-61581.1/T)}$	Hwang and Mebel (2003)
R1014	$N_2 + CH$	$\longrightarrow$	$N + HCN$	$6.11 \times 10^{-17}\, T^{1.42}\, e^{(-10428/T)}$	Miller and Walch (1997)
R1016	$N_2 + {}^3CH_2$	$\longrightarrow$	$NH + HCN$	$1.66 \times 10^{-11}\, e^{(-37237/T)}$	Smith *et al.* (2000)
R1018	$N_2 + {}^1CH_2$	$\longrightarrow$	$NH + HCN$	$1.66 \times 10^{-13}\, e^{(-32708/T)}$	Smith *et al.* (2000)
R1020	NH	$\xrightarrow{M}$	$N + H$	$k_0 = 4.4 \times 10^{-10}\, e^{(-38000/T)}$	Mertens *et al.* (1989)
R1022	$NH + CH$	$\longrightarrow$	$HCN + H$	1.0×10^{-10}	Estimate
R1024	$NH + {}^3CH_2$	$\longrightarrow$	$N + CH_3$	2.0×10^{-11}	Estimate
R1026	$NH + {}^3CH_2$	$\longrightarrow$	$H_2CN + H$	6.0×10^{-11}	Estimate
R1028	$NH + CH_3$	$\longrightarrow$	$N + CH_4$	1.0×10^{-13}	Estimate
R1030	$NH + CH_3$	$\longrightarrow$	$CH_2NH + H$	6.6×10^{-11}	Dean and Bozzelli (2000)
R1032	$NH + C_2H_3$	$\longrightarrow$	$N + C_2H_4$	1.0×10^{-11}	Estimate
R1034	$NH + C_2H_3$	$\longrightarrow$	$NH_2 + C_2H_2$	3.0×10^{-11}	Estimate
R1036	$NH + C_2H_3$	$\longrightarrow$	$CH_3CN + H$	5.0×10^{-11}	Estimate
R1038	$NH + C_2H_5$	$\longrightarrow$	$N + C_2H_6$	1.0×10^{-11}	Estimate
R1040	$NH + C_2H_5$	$\longrightarrow$	$NH_2 + C_2H_4$	3.0×10^{-11}	Estimate
R1042	$NH + C_2H_6$	$\longrightarrow$	$NH_2 + C_2H_5$	$7.42 \times 10^{-13}\, e^{(-4830/T)}$	Rohrig and Wagner (1994a)
R1044	$NH + O$	$\longrightarrow$	$N + OH$	$2.8 \times 10^{-16}\, T^{1.5}\, e^{(-1695/T)}$	Dean and Bozzelli (2000)
R1046	$NH + O$	$\longrightarrow$	$NO + H$	1.16×10^{-10}	Cohen and Westberg (1991)
R1048	$NH + O_2$	$\longrightarrow$	$NO + OH$	$7.47 \times 10^{-16}\, T^{0.79}\, e^{(-601/T)}$	Romming and Wagner (1996)
R1050	$NH + O_2$	$\longrightarrow$	$HNO + O$	$7.65 \times 10^{-19}\, T^{2}\, e^{(-3270/T)}$	Miller and Melius (1992a)
R1052	$NH + O_2$	$\longrightarrow$	$NO_2 + H$	$3.8 \times 10^{-14}\, e^{(-1250/T)}$	Dean and Bozzelli (2000)
R1054	$NH + OH$	$\longrightarrow$	$N + H_2O$	$3.3 \times 10^{-15}\, T^{1.2}$	Cohen and Westberg (1991)
R1056	$NH + OH$	$\longrightarrow$	$HNO + H$	3.3×10^{-11}	Cohen and Westberg (1991)
R1058	$NH + CO_2$	$\longrightarrow$	$HNO + CO$	$1.66 \times 10^{-11}\, e^{(-7220/T)}$	Rohrig and Wagner (1994b)

	Reaction	Rate constant	Reference
R1060	$2\,NH \longrightarrow N + NH_2$	$9.4 \times 10^{-25}\,T^{3.88}\,e^{(-172/T)}$	Klippenstein *et al.* (2009)
R1062	$2\,NH \longrightarrow N_2H + H$	$1.04 \times 10^{-10}\,T^{-0.036}\,e^{(81/T)}$	Based on Klippenstein *et al.* (2009)
R1064	$NH + NH_2 \longrightarrow NH_3 + N$	$1.59 \times 10^{-20}\,T^{2.46}\,e^{(-54/T)}$	Klippenstein *et al.* (2009)
R1066	$NH + NH_2 \longrightarrow N_2H_2 + H$	$7.07 \times 10^{-10}\,T^{-0.272}\,e^{(39/T)}$	Klippenstein *et al.* (2009)
R1068	$NH + N_2H_2 \longrightarrow N_2H + NH_2$	$4.0 \times 10^{-18}\,T^{2}\,e^{(600/T)}$	Dean and Bozzelli (2000)
R1070	$NH + H_2CN \longrightarrow NH_2 + HCN$	3.0×10^{-11}	Estimate
R1072	$NH + CH_3NH \longrightarrow N + CH_3NH_2$	1.0×10^{-11}	Estimate
R1074	$NH + CH_3NH \longrightarrow NH_2 + CH_2NH$	3.0×10^{-11}	Estimate
R1076	$NH + CH_3NH \longrightarrow NH_3 + H_2CN$	$2.0 \times 10^{-13}\,e^{(-2000/T)}$	Estimate
R1078	$NH + CH_2NH_2 \longrightarrow N + CH_3NH_2$	1.0×10^{-11}	Estimate
R1080	$NH + CH_2NH_2 \longrightarrow NH_2 + CH_2NH$	3.0×10^{-11}	Estimate
R1082	$NH + CH_2NH_2 \longrightarrow NH_3 + H_2CN$	$2.0 \times 10^{-13}\,e^{(-2000/T)}$	Estimate
R1084	$NH + CH_2CN \longrightarrow N + CH_3CN$	1.0×10^{-11}	Estimate
R1086	$NH + NO \longrightarrow N_2 + OH$	$1.01 \times 10^{-10}\,T^{-0.5}\,e^{(-60/T)}$	Bozzelli *et al.* (1994)
R1088	$NH + NO \longrightarrow N_2H + O$	$2.82 \times 10^{-10}\,T^{-0.2}\,e^{(-6140/T)}$	Dean and Bozzelli (2000)
R1090	$NH + NO \longrightarrow N_2O + H$	3.83×10^{-11}	Est., Quandt and Hershberger (1995)
R1092	$NH_2 \xrightarrow{M} NH + H$	$k_0 = 5.25 \times 10^{-1}\,T^{-2}\,e^{(-46024/T)}$	Hanson and Salimian (1984)
R1094	$NH_2 + H \longrightarrow NH + H_2$	$1.05 \times 10^{-10}\,e^{(-4450/T)}$	Rohrig and Wagner (1994b)
R1096	$NH_2 + H \xrightarrow{M} NH_3$	$k_0 = 5.33 \times 10^{-25}\,T^{-2}$	Est. based on Pagsberg *et al.* (1979)
		$k_\infty = 2.66 \times 10^{-11}$	Pagsberg *et al.* (1979)
R1098	$NH_2 + CH \longrightarrow H_2CN + H$	8.00×10^{-11}	Estimate
R1100	$NH_2 + CH \xrightarrow{M} CH_2NH$	$k_0 = 1.0 \times 10^{-32}$	Estimate
		$k_\infty = 1.0 \times 10^{-15}$	Estimate
R1102	$NH_2 + {}^3CH_2 \longrightarrow CH_2NH + H$	2.0×10^{-11}	Estimate
R1104	$NH_2 + CH_3 \longrightarrow CH_2NH_2 + H$	$2.32 \times 10^{-10}\,T^{-0.43}\,e^{(-5590/T)}$	Dean and Bozzelli (2000)
R1106	$NH_2 + CH_3 \longrightarrow CH_2NH + H_2$	$7.97 \times 10^{-13}\,T^{-0.2}\,e^{(-9765/T)}$	Dean and Bozzelli (2000)
R1108	$NH_2 + CH_3 \longrightarrow NH + CH_4$	$4.65 \times 10^{-18}\,T^{1.94}\,e^{(-4635/T)}$	Dean and Bozzelli (2000)
R1110	$NH_2 + CH_3 \longrightarrow NH_3 + {}^3CH_2$	$2.66 \times 10^{-18}\,T^{1.87}\,e^{(-3810/T)}$	Dean and Bozzelli (2000)
R1112	$NH_2 + CH_3 \xrightarrow{M} CH_3NH_2$	$k_0 = 2.01 \times 10^{-18}\,T^{-3.85}$	Jodkowski *et al.* (1995) divided
		$k_\infty = 3.97 \times 10^{-12}\,T^{0.42}$	by 3; see Moses *et al.* (2010)
R1114	$NH_2 + CH_4 \longrightarrow NH_3 + CH_3$	$8.3 \times 10^{-13}\,e^{(-5284/T)}$	Demissy and Lesclaux (1980)
R1116	$NH_2 + C_2H \longrightarrow NH + C_2H_2$	4.0×10^{-11}	Estimate
R1118	$NH_2 + C_2H_3 \longrightarrow NH_3 + C_2H_2$	5.0×10^{-11}	Estimate
R1120	$NH_2 + C_2H_5 \longrightarrow NH_3 + C_2H_4$	3.15×10^{-11}	based on Demissy and Lesclaux (1982)
R1122	$NH_2 + C_2H_6 \longrightarrow NH_3 + C_2H_5$	$7.21 \times 10^{-21}\,T^{2.636}\,e^{(-2630/T)}$	Based on Demissy and Lesclaux (1980)
			and Hennig and Wagner (1995)
R1124	$NH_2 + C_3H_8 \longrightarrow NH_3 + C_3H_7$	$7.47 \times 10^{-13}\,e^{(-3095/T)}$	Demissy and Lesclaux (1980)
R1126	$NH_2 + O \longrightarrow HNO + H$	6.0×10^{-11}	Adamson *et al.* (1994)
R1128	$NH_2 + O \longrightarrow NH + OH$	5.0×10^{-12}	Adamson *et al.* (1994)
R1130	$NH_2 + O_2 \longrightarrow HNO + OH$	$1.03 \times 10^{-16}\,T^{1.23}\,e^{(-17700/T)}$	Dean and Bozzelli (2000)
R1132	$NH_2 + OH \longrightarrow NH + H_2O$	$1.5 \times 10^{-16}\,T^{1.5}\,e^{(230/T)}$	Cohen and Westberg (1991)
R1134	$NH_2 + HCO \longrightarrow NH_3 + CO$	5.0×10^{-11}	Estimate

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R1136	$2\,NH_2 \longrightarrow NH_3 + NH$	$9.36 \times 10^{-24}\,T^{3.53}\,e^{(-278/T)}$	Klippenstein *et al.* (2009)
R1138	$2\,NH_2 \longrightarrow H_2NN + H_2$	$1.19 \times 10^{-19}\,T^{1.88}\,e^{(-4430/T)}$	Klippenstein *et al.* (2009)
R1140	$2\,NH_2 \longrightarrow N_2H_2 + H_2$	$2.89 \times 10^{-16}\,T^{1.02}\,e^{(-5930/T)}$	Klippenstein *et al.* (2009)
R1142	$2\,NH_2 \xrightarrow{M} N_2H_4$	$k_0 = 4.48 \times 10^{-14}\,T^{-5.49}\,e^{(-1000/T)}$	Klippenstein *et al.* (2009)
		$k_\infty = 9.33 \times 10^{-10}\,T^{-0.414}\,e^{(-33/T)}$	Klippenstein *et al.* (2009)
		$F_c = 0.31$	Klippenstein *et al.* (2009)
R1144	$NH_2 + NH_3 \longrightarrow N_2H_3 + H_2$	$4.0 \times 10^{-9}\,e^{(-34200/T)}$	Dean *et al.* (1984)
R1146	$NH_2 + N_2H \longrightarrow N_2 + NH_3$	$1.53 \times 10^{-18}\,T^{1.94}\,e^{(580/T)}$	Dean and Bozzelli (2000)
R1148	$NH_2 + N_2H_2 \longrightarrow NH_3 + N_2H$	$3.0 \times 10^{-18}\,T^{1.94}\,e^{(580/T)}$	Dean and Bozzelli (2000)
R1150	$NH_2 + H_2NN \longrightarrow NH_3 + N_2H$	$3.0 \times 10^{-18}\,T^{1.94}\,e^{(580/T)}$	Dean and Bozzelli (2000)
R1152	$NH_2 + N_2H_3 \longrightarrow NH_3 + H_2NN$	5.0×10^{-11}	Dean and Bozzelli (2000)
R1154	$NH_2 + N_2H_3 \longrightarrow NH_3 + N_2H_2$	$1.53 \times 10^{-18}\,T^{1.94}\,e^{(580/T)}$	Dean and Bozzelli (2000)
R1156	$NH_2 + N_2H_4 \longrightarrow NH_3 + N_2H_3$	$4.298 \times 10^{-17}\,T^{1.94}\,e^{(-820/T)}$	Moses *et al.* (2010)
R1158	$NH_2 + H_2CN \longrightarrow NH_3 + HCN$	$2.29 \times 10^{-8}\,T^{-1.06}\,e^{(-60.8/T)}$	Yelle *et al.* (2010)
R1160	$NH_2 + CH_2NH \longrightarrow NH_3 + H_2CN$	$1.5 \times 10^{-18}\,T^{1.94}\,e^{(-2235/T)}$	Dean and Bozzelli (2000)
R1162	$NH_2 + CH_3NH \longrightarrow NH_3 + CH_2NH$	4.0×10^{-11}	Estimate
R1164	$NH_2 + CH_2NH_2 \longrightarrow NH_3 + CH_2NH$	4.0×10^{-11}	Estimate
R1166	$NH_2 + CH_3NH_2 \longrightarrow NH_3 + CH_2NH_2$	$4.65 \times 10^{-18}\,T^{1.94}\,e^{(-2765/T)}$	Dean and Bozzelli (2000)
R1168	$NH_2 + CH_3NH_2 \longrightarrow NH_3 + CH_3NH$	$3.0 \times 10^{-18}\,T^{1.94}\,e^{(-3595/T)}$	Dean and Bozzelli (2000)
R1170	$NH_2 + CH_2CN \longrightarrow HCN + CH_2NH$	5.0×10^{-12}	Estimate
R1172	$NH_2 + NO \longrightarrow N_2 + H_2O$	$7.8 \times 10^{-12}\,T^{-0.25}\,e^{(605/T)}$	Dean and Bozzelli (2000)
R1174	$NH_2 + NO_2 \longrightarrow N_2O + H_2O$	$2.56 \times 10^{-8}\,T^{-1.44}\,e^{(-135/T)}$	Park and Lin (1997)
R1176	$NH_2 + HNO \longrightarrow NO + NH_3$	$1.5 \times 10^{-18}\,T^{1.94}\,e^{(580/T)}$	Dean and Bozzelli (2000)
R1178	$NH_3 + H \longrightarrow NH_2 + H_2$	$7.77 \times 10^{-24}\,T^{3.93}\,e^{(-4063/T)}$	Espinosa-Garcia and Corchado (1994)
R1180	$NH_3 + CH \longrightarrow CH_2NH + H$	$4.13 \times 10^{-9}\,T^{-0.56}\,e^{(-28/T)}$	Bocherel *et al.* (1996)
R1182	$NH_3 + O \longrightarrow NH_2 + OH$	$1.83 \times 10^{-18}\,T^{2.1}\,e^{(-2620/T)}$	Cohen and Westberg (1991)
R1184	$NH_3 + O(^1D) \longrightarrow NH_2 + OH$	2.5×10^{-10}	DeMore *et al.* (1987)
R1186	$NH_3 + OH \longrightarrow NH_2 + H_2O$	$8.31 \times 10^{-17}\,T^{1.6}\,e^{(-480/T)}$	Cohen and Westberg (1991)
R1188	$NH_3 + CN \longrightarrow NH_2 + HCN$	$1.52 \times 10^{-11}\,e^{(180/T)}$	Sims and Smith (1988)
R1190	$NH_3 + NCO \longrightarrow NH_2 + HNCO$	$4.65 \times 10^{-20}\,T^{2.48}\,e^{(-495/T)}$	Becker *et al.* (1997)
R1192	$N_2H \longrightarrow N_2 + H$	3.0×10^{8}	Est., Dean and Bozzelli (2000)
		$1.3 \times 10^{15}\,T^{-0.53}\,e^{(-3404/T)}$	Test case: Caridade *et al.* (2007)
		$1.115 \times 10^{11}\,T^{0.808}\,e^{(-2429.8/T)}$	Test case: fit to Hwang and Mebel (2003)
R1194	$N_2H \xrightarrow{M} N_2 + H$	$1.66 \times 10^{-11}\,T^{0.5}\,e^{(-1540/T)}$	Dean and Bozzelli (2000)
		0.	Test case: Caridade *et al.* (2007)
		0.	Test case: Hwang and Mebel (2003)
R1196	$N_2H + H \longrightarrow N_2 + H_2$	$4.0 \times 10^{-16}\,T^{1.5}\,e^{(450/T)}$	Dean and Bozzelli (2000)
R1198	$N_2H + CH_3 \longrightarrow N_2 + CH_4$	8.0×10^{-11}	Estimate
R1200	$N_2H + O \longrightarrow N_2 + OH$	$2.82 \times 10^{-8}\,T^{-1.23}\,e^{(-250/T)}$	Dean and Bozzelli (2000)
R1202	$N_2H + OH \longrightarrow N_2 + H_2O$	$3.98 \times 10^{-2}\,T^{-2.88}\,e^{(-1230/T)}$	Dean and Bozzelli (2000)
R1204	$N_2H + OH \longrightarrow NH_2 + NO$	$1.52 \times 10^{-10}\,T^{-0.645}$	Estimate
R1206	$N_2H + NO \longrightarrow N_2 + HNO$	$1.99 \times 10^{-18}\,T^{2}\,e^{(600/T)}$	Dean and Bozzelli (2000)
R1208	$N_2H + N_2H_3 \longrightarrow 2\,N_2H_2$	$1.66 \times 10^{-11}\,e^{(-2015/T)}$	Konnov and De Ruyck (2001)
R1210	$N_2H_2 \longrightarrow N_2H + H$	$1.6 \times 10^{37}\,T^{-7.94}\,e^{(-35600/T)}$	Dean and Bozzelli (2000)
R1212	$N_2H_2 \longrightarrow H_2NN$	$2.00 \times 10^{41}\,T^{-9.38}\,e^{(-34450/T)}$	Dean and Bozzelli (2000)

TABLE S1 (*cont.*)

	Reaction			Rate constant	Reference
R1214	$N_2H_2 + H$	$\longrightarrow$	$N_2H + H_2$	$7.97 \times 10^{-16}\, T^{1.5}\, e^{(-795/T)}$	Dean and Bozzelli (2000)
R1216	$N_2H_2 + CH_3$	$\longrightarrow$	$N_2H + CH_4$	$2.66 \times 10^{-18}\, T^{1.87}\, e^{(-1495/T)}$	Dean and Bozzelli (2000)
R1218	$N_2H_2 + O$	$\longrightarrow$	$N_2H + OH$	$5.5 \times 10^{-16}\, T^{1.5}\, e^{(-250/T)}$	Dean and Bozzelli (2000)
R1220	$N_2H_2 + O$	$\longrightarrow$	$NH_2 + NO$	1.66×10^{-11}	Est., Miller and Bowman (1989)
R1222	$N_2H_2 + OH$	$\longrightarrow$	$N_2H + H_2O$	$4.0 \times 10^{-18}\, T^{2}\, e^{(600/T)}$	Dean and Bozzelli (2000)
R1224	H_2NN	$\longrightarrow$	$N_2H + H$	$3.2 \times 10^{31}\, T^{-6.22}\, e^{(-26330/T)}$	Dean and Bozzelli (2000)
R1226	H_2NN	$\longrightarrow$	N_2H_2	$4.076 \times 10^{11}\, T^{0.716}\, e^{(-23004.6/T)}$	Hwang and Mebel (2003)
R1228	$H_2NN + H$	$\longrightarrow$	$N_2H_2 + H$	$3.0 \times 10^{-14}\, T^{0.97}\, e^{(-2250/T)}$	Dean and Bozzelli (2000)
R1230	$H_2NN + H$	$\longrightarrow$	$N_2H + H_2$	$7.97 \times 10^{-16}\, T^{1.5}\, e^{(450/T)}$	Dean and Bozzelli (2000)
R1232	$H_2NN + H_2$	$\longrightarrow$	N_2H_4	$3.47 \times 10^{-19}\, T^{2.27}\, e^{(-13814/T)}$	Hwang and Mebel (2003)
R1234	N_2H_3	$\xrightarrow{M}$	$N_2H_2 + H$	$k_0 = 3.49 \times 10^{38}\, T^{-13.13}\, e^{(-36825.4/T)}$	based on Dean and Bozzelli (2000)
				$k_\infty = 7.95 \times 10^{13}\, e^{(-27463/T)}$	Hwang and Mebel (2003)
R1236	$N_2H_3 + H$	$\longrightarrow$	$2\,NH_2$	1.5×10^{-10}	Estimate
R1238	$N_2H_3 + H$	$\longrightarrow$	$N_2H_2 + H_2$	$4.0 \times 10^{-16}\, T^{1.5}$	Dean and Bozzelli (2000)
R1240	$N_2H_3 + H$	$\longrightarrow$	$H_2NN + H_2$	$1.0 \times 10^{-17}\, T^{1.5}$	Estimate
R1242	$N_2H_3 + CH_3$	$\longrightarrow$	$N_2H_2 + CH_4$	$1.36 \times 10^{-18}\, T^{1.87}\, e^{(-915/T)}$	Dean and Bozzelli (2000)
R1244	$N_2H_3 + O$	$\longrightarrow$	$NH_2 + HNO$	1.0×10^{-10}	Estimate
R1246	$N_2H_3 + O$	$\longrightarrow$	$N_2H_2 + OH$	$2.82 \times 10^{-16}\, T^{1.5}\, e^{(325/T)}$	Dean and Bozzelli (2000)
R1248	$N_2H_3 + OH$	$\longrightarrow$	$N_2H_2 + H_2O$	$2.0 \times 10^{-18}\, T^{2}\, e^{(600/T)}$	Dean and Bozzelli (2000)
R1250	$2\,N_2H_3$	$\longrightarrow$	$N_2H_2 + N_2H_4$	2.0×10^{-11}	Konnov and De Ruyck (2001)
R1252	$2\,N_2H_3$	$\longrightarrow$	$2\,NH_3 + N_2$	5.0×10^{-12}	Est., Konnov and De Ruyck (2001) and Stief (1970)
R1254	N_2H_4	$\longrightarrow$	$H_2NN + H_2$	$k_\infty = 5.3 \times 10^{39}\, T^{-8.35}\, e^{(-34880/T)}$	Dean and Bozzelli (2000)
R1256	$N_2H_4 + H$	$\longrightarrow$	$N_2H_3 + H_2$	$1.17 \times 10^{-11}\, e^{(-1260/T)}$	Vaghjiani (1995)
R1258	$N_2H_4 + H$	$\longrightarrow$	$NH_2 + NH_3$	$3.84 \times 10^{-16}\, T^{1.42}\, e^{(-4128/T)}$	Hwang and Mebel (2003)
R1260	$N_2H_4 + CH_3$	$\longrightarrow$	$N_2H_3 + CH_4$	$1.66 \times 10^{-13}\, e^{(-2516/T)}$	Gray and Thynne (1965)
R1262	$N_2H_4 + O$	$\longrightarrow$	$N_2H_3 + OH$	$7.35 \times 10^{-13}\, e^{(640/T)}$	Vaghjiani (1996)
R1264	$N_2H_4 + OH$	$\longrightarrow$	$N_2H_3 + H_2O$	3.6×10^{-11}	Vaghjiani (1996)
R1266	$N_2H_4 + CN$	$\longrightarrow$	$N_2H_3 + HCN$	$1.83 \times 10^{-8}\, T^{-1.14}$	Estimate based on CN + NH_3
R1268	$CN + H_2$	$\longrightarrow$	$HCN + H$	$3.2 \times 10^{-20}\, T^{2.87}\, e^{(-820/T)}$	Baulch *et al.* (1994)
R1270	$CN + CH_3$	$\longrightarrow$	$HCN + {}^3CH_2$	3.0×10^{-11}	Estimate
R1272	$CN + CH_4$	$\longrightarrow$	$HCN + CH_3$	$5.15 \times 10^{-16}\, T^{1.53}\, e^{(-504/T)}$	Yang *et al.* (1993)
R1274	$CN + C_2H$	$\xrightarrow{M}$	HC_3N	$k_0 = 1.0 \times 10^{-30}$	Estimate
				$k_\infty = 1.29 \times 10^{-11}$	Millar *et al.* (1991)
R1276	$CN + C_2H_2$	$\longrightarrow$	$HC_3N + H$	$5.26 \times 10^{-9}\, T^{-0.52}\, e^{(-19/T)}$	Sims *et al.* (1993)
R1278	$CN + C_2H_4$	$\longrightarrow$	$HCN + C_2H_3$	$1.1 \times 10^{-10}\, e^{(172/T)}$	Monks *et al.* (1993)
R1280	$CN + C_2H_4$	$\longrightarrow$	$C_2H_3CN + H$	$2.8 \times 10^{-11}\, e^{(172/T)}$	Monks *et al.* (1993)
R1282	$CN + C_2H_6$	$\longrightarrow$	$HCN + C_2H_5$	$5.91 \times 10^{-12}\, T^{0.22}\, e^{(58/T)}$	Sims *et al.* (1993)
R1284	$CN + CH_3C_2H$	$\longrightarrow$	$HCN + C_3H_3$	2.1×10^{-10}	Sayah *et al.* (1988)
R1286	$CN + C_3H_8$	$\longrightarrow$	$HCN + C_3H_7$	$2.44 \times 10^{-14}\, T^{1.19}\, e^{(378/T)}$	Yang *et al.* (1992)
R1288	$CN + O$	$\longrightarrow$	$N + CO$	$3.4 \times 10^{-11}\, e^{(-210/T)}$	Tsang (1992)
R1290	$CN + O_2$	$\longrightarrow$	$NCO + O$	$1.71 \times 10^{-9}\, T^{-0.76}\, e^{(-7/T)}$	Sims *et al.* (1992)
R1292	$CN + OH$	$\longrightarrow$	$NCO + H$	7.0×10^{-11}	Tsang (1992)
R1294	$CN + CO_2$	$\longrightarrow$	$NCO + CO$	$6.09 \times 10^{-18}\, T^{2.16}\, e^{(-13530/T)}$	Wang *et al.* (1991)
R1296	$CN + H_2CO$	$\longrightarrow$	$HCN + HCO$	$2.82 \times 10^{-21}\, T^{2.72}\, e^{(718/T)}$	Feng *et al.* (1997)

TABLE S1 (*cont.*)

	Reaction			Rate constant	Reference
R1298	$CN + CH_3OH$	$\longrightarrow$	$HCN + CH_2OH$	1.2×10^{-10}	Sayah *et al.* (1988)
R1300	$CN + H_2CN$	$\longrightarrow$	$2\,HCN$	$5.26 \times 10^{-9}\,T^{-0.52}\,e^{(-19/T)}$	Estimate based on CN + C_2H_2
R1302	$CN + CH_3CN$	$\longrightarrow$	$HCN + CH_2CN$	$6.46 \times 10^{-11}\,e^{(-1190/T)}$	Zabarnick and Lin (1989)
R1304	$CN + HC_3N$	$\longrightarrow$	$HCN + C_3N$	8.5×10^{-12}	Halpern *et al.* (1989)
R1306	$CN + NO_2$	$\longrightarrow$	$NCO + NO$	$3.98 \times 10^{-11}\,e^{(186/T)}$	Wang *et al.* (1989)
R1308	$CN + HNCO$	$\longrightarrow$	$HCN + NCO$	2.5×10^{-11}	Tsang (1992)
R1310	HCN	$\xrightarrow{M}$	$CN + H$	$k_0 = 593\,T^{-2.6}\,e^{(-62845/T)}$	Tsang and Herron (1991)
				$k_\infty = 8.3 \times 10^{17}\,T^{-0.93}\,e^{(-62294/T)}$	Tsang and Herron (1991)
R1312	$HCN + H$	$\xrightarrow{M}$	H_2CN	$k_0 = 4.4 \times 10^{-24}\,T^{-2.73}\,e^{(-3855/T)}$	Tsang and Herron (1991)
				$k_\infty = 5.5 \times 10^{-11}\,e^{(-2438/T)}$	Tsang and Herron (1991)
R1314	$HCN + CH$	$\xrightarrow{M}$	CH_2CN	$k_0 = 1.0 \times 10^{-30}$	Estimate
				$k_\infty = 5.0 \times 10^{-11}\,e^{(500/T)}$	Zabarnick *et al.* (1991)
R1316	$HCN + {}^3CH_2$	$\longrightarrow$	$CH_2CN + H$	$5.0 \times 10^{-11}\,e^{(-1400/T)}$	Estimate
R1318	$HCN + C_2H$	$\longrightarrow$	$HC_3N + H$	$5.3 \times 10^{-12}\,e^{(-769/T)}$	Hoobler and Leone (1997)
R1320	$HCN + C_2H_3$	$\longrightarrow$	$C_2H_3CN + H$	$1.0 \times 10^{-12}\,e^{(-900/T)}$	Monks *et al.* (1993)
R1322	$HCN + O$	$\longrightarrow$	$NCO + H$	$3.3 \times 10^{-16}\,T^{1.47}\,e^{(-3774/T)}$	Perry and Melius (1985)
R1324	$HCN + O$	$\longrightarrow$	$NH + CO$	$9.0 \times 10^{-16}\,T^{1.21}\,e^{(-3824/T)}$	Perry and Melius (1985)
R1326	$HCN + O$	$\longrightarrow$	$CN + OH$	$4.5 \times 10^{-15}\,T^{1.58}\,e^{(-13386/T)}$	Perry and Melius (1985)
R1328	$HCN + OH$	$\longrightarrow$	$HNCO + H$	$9.3 \times 10^{-30}\,T^{4.71}\,e^{(248/T)}$	Tsang and Herron (1991)
R1330	$HCN + OH$	$\longrightarrow$	$CN + H_2O$	$3.6 \times 10^{-17}\,T^{1.5}\,e^{(-3887/T)}$	Tsang and Herron (1991)
R1332	$H_2CN + H$	$\longrightarrow$	$HCN + H_2$	7.0×10^{-11}	Nesbitt *et al.* (1990)
R1334	$H_2CN + H$	$\xrightarrow{M}$	CH_2NH	$k_0 = 5.0 \times 10^{-25}\,T^{-1}$	Estimate
				$k_\infty = 5.0 \times 10^{-11}$	Estimate
R1336	$H_2CN + H_2$	$\longrightarrow$	$CH_2NH + H$	$4.4 \times 10^{-22}\,T^{3.051}\,e^{(-7990/T)}$	Klippenstein (2010)
R1338	$H_2CN + CH_3$	$\longrightarrow$	$HCN + CH_4$	$1.35 \times 10^{-18}\,T^{1.87}\,e^{(560/T)}$	Dean and Bozzelli (2000)
R1340	$H_2CN + C_2H_3$	$\longrightarrow$	$CH_2NH + C_2H_2$	2.4×10^{-11}	Estimate
R1342	$H_2CN + C_2H_5$	$\longrightarrow$	$CH_2NH + C_2H_4$	3.0×10^{-11}	Estimate
R1344	$H_2CN + O$	$\longrightarrow$	$HCN + OH$	$2.8 \times 10^{-16}\,T^{1.5}\,e^{(450/T)}$	Dean and Bozzelli (2000)
R1346	$H_2CN + O$	$\longrightarrow$	$HNCO + H$	1.0×10^{-10}	Dean and Bozzelli (2000)
R1348	$H_2CN + OH$	$\longrightarrow$	$HCN + H_2O$	$2.0 \times 10^{-18}\,T^{2}\,e^{(600/T)}$	Dean and Bozzelli (2000)
R1350	$2\,H_2CN$	$\longrightarrow$	$HCN + CH_2NH$	7.7×10^{-12}	Moses *et al.* (2010)
R1352	$CH_2NH + CH_3$	$\longrightarrow$	$H_2CN + CH_4$	$1.36 \times 10^{-18}\,T^{1.87}\,e^{(-3585/T)}$	Dean and Bozzelli (2000)
R1354	$CH_2NH + C_2H$	$\longrightarrow$	$H_2CN + C_2H_2$	$5.0 \times 10^{-16}\,T^{1.5}\,e^{(-1000/T)}$	Estimate
R1356	$CH_2NH + C_2H_3$	$\longrightarrow$	$H_2CN + C_2H_4$	$5.0 \times 10^{-15}\,T^{1.6}\,e^{(-1800/T)}$	Estimate
R1358	$CH_2NH + C_2H_5$	$\longrightarrow$	$H_2CN + C_2H_6$	$1.0 \times 10^{-14}\,e^{(-2000/T)}$	Estimate
R1360	$CH_2NH + O$	$\longrightarrow$	$NH + H_2CO$	$2.82 \times 10^{-18}\,T^{2.08}$	Dean and Bozzelli (2000)
R1362	$CH_2NH + O$	$\longrightarrow$	$H_2CN + OH$	$2.82 \times 10^{-16}\,T^{1.5}\,e^{(-2330/T)}$	Dean and Bozzelli (2000)
R1364	$CH_2NH + OH$	$\longrightarrow$	$H_2CN + H_2O$	$2.0 \times 10^{-18}\,T^{2}\,e^{(45/T)}$	Dean and Bozzelli (2000)
R1366	CH_3NH	$\longrightarrow$	$CH_2NH + H$	$1.3 \times 10^{42}\,T^{-9.24}\,e^{(-20805/T)}$	Dean and Bozzelli (2000)
R1368	$CH_3NH + H$	$\longrightarrow$	$NH_2 + CH_3$	1.0×10^{-10}	Estimate
R1370	$CH_3NH + H$	$\longrightarrow$	$CH_2NH + H_2$	$1.2 \times 10^{-15}\,T^{1.5}\,e^{(450/T)}$	Dean and Bozzelli (2000)
R1372	$CH_3NH + H$	$\xrightarrow{M}$	CH_3NH_2	$k_0 = 3.0 \times 10^{-26}\,T^{-1}$	Estimate
				$k_\infty = 5.0 \times 10^{-11}$	Estimate
R1374	$CH_3NH + CH_3$	$\longrightarrow$	$CH_2NH + CH_4$	$4.0 \times 10^{-18}\,T^{1.87}\,e^{(560/T)}$	Dean and Bozzelli (2000)
R1376	$CH_3NH + O$	$\longrightarrow$	$CH_2NH + OH$	$8.3 \times 10^{-16}\,T^{1.5}\,e^{(450/T)}$	Dean and Bozzelli (2000)
R1378	$CH_3NH + OH$	$\longrightarrow$	$CH_2NH + H_2O$	$6.0 \times 10^{-18}\,T^{2}\,e^{(600/T)}$	Dean and Bozzelli (2000)

TABLE S1 (*cont.*)

	Reaction	Rate constant	Reference
R1380	$CH_2NH_2 \longrightarrow CH_2NH + H$	$3.2 \times 10^{46}\, T^{-9.95}\, e^{(-26940/T)}$	Dean and Bozzelli (2000)
R1382	$CH_2NH_2 + H \longrightarrow CH_2NH + H_2$	$8.0 \times 10^{-16}\, T^{1.5}\, e^{(450/T)}$	Dean and Bozzelli (2000)
R1384	$CH_2NH_2 + CH_3 \longrightarrow CH_2NH + CH_4$	$2.66 \times 10^{-18}\, T^{1.87}\, e^{(315/T)}$	Dean and Bozzelli (2000)
R1386	$CH_2NH_2 + CH_3 \longrightarrow NH_2 + C_2H_5$	$3.32 \times 10^{-11}\, e^{(-1360/T)}$	Dean and Bozzelli (2000)
R1388	$CH_2NH_2 + O \longrightarrow NH_2 + H_2CO$	1.16×10^{-10}	Dean and Bozzelli (2000)
R1390	$CH_2NH_2 + O \longrightarrow CH_2NH + OH$	$5.5 \times 10^{-16}\, T^{1.5}\, e^{(450/T)}$	Dean and Bozzelli (2000)
R1392	$CH_2NH_2 + OH \longrightarrow NH_2 + CH_2OH$	6.6×10^{-11}	Dean and Bozzelli (2000)
R1394	$CH_2NH_2 + OH \longrightarrow CH_2NH + H_2O$	$4.0 \times 10^{-18}\, T^{2}\, e^{(600/T)}$	Dean and Bozzelli (2000)
R1396	$CH_3NH_2 + H \longrightarrow CH_2NH_2 + H_2$	$9.3 \times 10^{-16}\, T^{1.5}\, e^{(-2750/T)}$	Dean and Bozzelli (2000)
R1398	$CH_3NH_2 + H \longrightarrow CH_3NH + H_2$	$8.0 \times 10^{-16}\, T^{1.5}\, e^{(-4885/T)}$	Dean and Bozzelli (2000)
R1400	$CH_3NH_2 + CH \longrightarrow NH_2 + C_2H_4$	$1.55 \times 10^{-10}\, e^{(170/T)}$	Est., Zabarnick *et al.* (1989)
R1402	$CH_3NH_2 + CH \longrightarrow NH_3 + C_2H_3$	$1.27 \times 10^{-10}\, e^{(170/T)}$	Est., Zabarnick *et al.* (1989)
R1404	$CH_3NH_2 + CH \longrightarrow CH_2NH + CH_3$	$9.3 \times 10^{-12}\, e^{(170/T)}$	Est., Zabarnick *et al.* (1989)
R1406	$CH_3NH_2 + CH \longrightarrow H_2CN + CH_4$	$9.3 \times 10^{-12}\, e^{(170/T)}$	Est., Zabarnick *et al.* (1989)
R1408	$CH_3NH_2 + CH \longrightarrow CH_3CN + H_2 + H$	$9.3 \times 10^{-12}\, e^{(170/T)}$	Est., Zabarnick *et al.* (1989)
R1410	$CH_3NH_2 + CH_3 \longrightarrow CH_2NH_2 + CH_4$	$2.5 \times 10^{-18}\, T^{1.87}\, e^{(-4615/T)}$	Dean and Bozzelli (2000)
R1412	$CH_3NH_2 + CH_3 \longrightarrow CH_3NH + CH_4$	$2.66 \times 10^{-18}\, T^{1.87}\, e^{(-4450/T)}$	Dean and Bozzelli (2000)
R1414	$CH_3NH_2 + OH \longrightarrow CH_2NH_2 + H_2O$	$6.0 \times 10^{-18}\, T^{2}\, e^{(120/T)}$	Dean and Bozzelli (2000)
R1416	$CH_3NH_2 + OH \longrightarrow CH_3NH + H_2O$	$4.0 \times 10^{-18}\, T^{2}\, e^{(225/T)}$	Dean and Bozzelli (2000)
R1418	$CH_3NH_2 + CN \longrightarrow CH_2NH_2 + HCN$	$5.2 \times 10^{-15}\, T^{1.26}\, e^{(208/T)}$	Estimate based on CN + C_2H_6
R1420	$CH_3NH_2 + CN \longrightarrow CH_3NH + HCN$	$5.2 \times 10^{-15}\, T^{1.26}\, e^{(208/T)}$	Estimate based on CN + C_2H_6
R1422	$CH_2CN + H \xrightarrow{M} CH_3CN$	$k_0 = 1.0 \times 10^{-29}$	Estimate
		$k_\infty = 1.0 \times 10^{-10}$	Estimate
R1424	$CH_2CN + C_2H_3 \longrightarrow CH_3CN + C_2H_2$	8.0×10^{-13}	Estimate
R1426	$CH_2CN + C_2H_5 \longrightarrow CH_3CN + C_2H_4$	2.0×10^{-12}	Estimate
R1428	$CH_3CN + H \longrightarrow HCN + CH_3$	$6.71 \times 10^{-11}\, e^{(-4844/T)}$	Williams and Fleming (1997)
R1430	$CH_3CN + H \longrightarrow CH_2CN + H_2$	$3.32 \times 10^{-12}\, e^{(-4277.8/T)}$	Williams and Fleming (1997)
R1432	$CH_3CN + CH_3 \longrightarrow CH_2CN + CH_4$	$1.0 \times 10^{-12}\, e^{(-3000/T)}$	Estimate
R1434	$CH_3CN + C_2H \longrightarrow HC_3N + CH_3$	$1.8 \times 10^{-11}\, e^{(-769/T)}$	Est., Hoobler and Leone (1997)
R1436	$CH_3CN + O \longrightarrow CH_2CN + OH$	$7.307 \times 10^{-12}\, e^{(-3291/T)}$	Williams and Fleming (1997)
R1438	$CH_3CN + O \longrightarrow NCO + CH_3$	$2.14 \times 10^{-12}\, e^{(-2100/T)}$	Est., Budge and Roscoe (1995)
R1440	$CH_3CN + OH \longrightarrow CH_2CN + H_2O$	$6.0 \times 10^{-13}\, e^{(-1050/T)}$	Est., DeMore *et al.* (1997)
R1442	$CH_3CN + OH \longrightarrow HNCO + CH_3$	$1.8 \times 10^{-13}\, e^{(-1050/T)}$	Est., DeMore *et al.* (1997)
R1444	$C_3N + H \xrightarrow{M} HC_3N$	$k_0 = 5.0 \times 10^{-22}\, T^{-1.5}$	Estimate
		$k_\infty = 2.0 \times 10^{-10}$	Estimate
R1446	$C_3N + H_2 \longrightarrow HC_3N + H$	$1.0 \times 10^{-11}\, e^{(-1300/T)}$	Estimate
R1448	$C_3N + CH_4 \longrightarrow HC_3N + CH_3$	$2.4 \times 10^{-11}\, e^{(-900/T)}$	Estimate
R1450	$C_3N + C_2H_6 \longrightarrow HC_3N + C_2H_5$	$3.0 \times 10^{-11}\, e^{(-400/T)}$	Estimate
R1452	$C_2H_2CN \xrightarrow{M} HC_3N + H$	$k_0 = 4.26 \times 10^{22}\, T^{-8.55}\, e^{(-32226)/T)}$	Kiefer *et al.* (1996)
R1454	$C_2H_2CN \xrightarrow{M} CN + C_2H_2$	$k_0 = 4.27 \times 10^{35}\, T^{-11.9}\, e^{(-42366)/T)}$	Kiefer *et al.* (1996)
R1456	$C_2H_2CN + H \longrightarrow C_2H_3CN$	1.0×10^{-10}	Estimate
R1458	$C_2H_2CN + H_2 \longrightarrow C_2H_3CN + H$	$8.0 \times 10^{-11}\, e^{(-2000/T)}$	Estimate
R1460	$C_2H_2CN + CH_4 \longrightarrow C_2H_3CN + CH_3$	$1.0 \times 10^{-10}\, e^{(-2300/T)}$	Estimate
R1462	$C_2H_2CN + H_2O \longrightarrow C_2H_3CN + OH$	$1.0 \times 10^{-10}\, e^{(-3000/T)}$	Estimate
R1464	$C_2H_3CN + {}^3CH_2 \longrightarrow CH_2CN + C_2H_3$	2.0×10^{-11}	Estimate

TABLE S1 (*cont.*)

	Reaction			Rate constant	Reference
R1466	$C_2H_3CN + C_2$	$\longrightarrow$	$HCN + C_4H_2$	2.0×10^{-10}	Est., Reisler *et al.* (1980)
R1468	$C_2H_3CN + C_2$	$\longrightarrow$	$HC_3N + C_2H_2$	2.4×10^{-10}	Est., Reisler *et al.* (1980)
R1470	$C_2H_3CN + O$	$\longrightarrow$	$HCN + H_2CCO$	$1.4 \times 10^{-11}\, e^{(-1000/T)}$	Estimate
R1472	$C_2H_3CN + OH$	$\longrightarrow$	$HCN + C_2H_3O$	8.5×10^{-12}	Grosjean and Williams (1992)
R1474	$NO + H$	$\xrightarrow{M}$	HNO	$k_0 = 6.46 \times 10^{-30}\, T^{-0.9}$	Clyne and Thrush (1962)
				$k_\infty = 5.4 \times 10^{-10}$	Forte (1981)
R1476	$NO + C$	$\longrightarrow$	$N + CO$	2.8×10^{-11}	Dean and Bozzelli (2000)
R1478	$NO + C$	$\longrightarrow$	$CN + O$	1.8×10^{-11}	Dean and Bozzelli (2000)
R1480	$NO + CH$	$\longrightarrow$	$HCN + O$	6.8×10^{-11}	Smith *et al.* (2000)
R1482	$NO + CH$	$\longrightarrow$	$NCO + H$	2.7×10^{-11}	Smith *et al.* (2000)
R1484	$NO + CH$	$\longrightarrow$	$N + HCO$	4.1×10^{-11}	Smith *et al.* (2000)
R1486	$NO + {}^1CH_2$	$\longrightarrow$	$HNCO + H$	$6.3 \times 10^{-11}\, T^{-0.4}\, e^{(-292/T)}$	Smith *et al.* (2000)
R1488	$NO + {}^1CH_2$	$\longrightarrow$	$HCN + OH$	$4.8 \times 10^{-10}\, T^{-0.7}\, e^{(-382/T)}$	Smith *et al.* (2000)
R1490	$NO + {}^3CH_2$	$\longrightarrow$	$HNCO + H$	$5.15 \times 10^{-7}\, T^{-1.4}\, e^{(-639/T)}$	Smith *et al.* (2000)
R1492	$NO + {}^3CH_2$	$\longrightarrow$	$HCN + OH$	$4.8 \times 10^{-10}\, T^{-0.7}\, e^{(-382/T)}$	Smith *et al.* (2000)
R1494	$NO + CH_3$	$\longrightarrow$	$HCN + H_2O$	$4.0 \times 10^{-12}\, e^{(-7900/T)}$	Hennig and Wagner (1994)
R1496	$NO + CH_3$	$\longrightarrow$	$H_2CN + OH$	$8.63 \times 10^{-12}\, e^{(-12200/T)}$	Hennig and Wagner (1994)
R1498	$NO + CH_3$	$\xrightarrow{M}$	CH_3NO	$k_0 = 8.9 \times 10^{-26}\, T^{-1.87}$	Davies *et al.* (1991)
				$k_\infty = 7.22 \times 10^{-8}\, T^{-1.58}$	Davies *et al.* (1991)
R1500	$NO + O$	$\xrightarrow{M}$	NO_2	$k_0 = 9.2 \times 10^{-28}\, T^{-1.6}$	Atkinson *et al.* (1992)
				$k_\infty = 5.4 \times 10^{-12}\, T^{0.3}$	Atkinson *et al.* (1992)
R1502	$NO + OH$	$\xrightarrow{M}$	HNO_2	$k_0 = 1.4 \times 10^{-24}\, T^{-2.51}\, e^{(34/T)}$	Tsang and Herron (1991)
				$k_\infty = 3.2 \times 10^{-11}$	Atkinson *et al.* (1992)
R1504	$NO + HCO$	$\longrightarrow$	$HNO + CO$	1.26×10^{-11}	Langford and Moore (1984)
R1506	$NO + CH_3O$	$\longrightarrow$	$HNO + H_2CO$	$2.2 \times 10^{-10}\, T^{-0.7}$	Atkinson *et al.* (1992)
R1508	$NO + HCCO$	$\longrightarrow$	$HCN + CO_2$	$1.0 \times 10^{-10}\, e^{(-350/T)}$	Boullart *et al.* (1994)
R1510	$NO + HO_2$	$\longrightarrow$	$NO_2 + OH$	$3.5 \times 10^{-12}\, e^{(240/T)}$	Lightfoot *et al.* (1992)
R1512	$NO + N_2O$	$\longrightarrow$	$N_2 + NO_2$	$8.74 \times 10^{-19}\, T^{2.23}\, e^{(-23292/T)}$	Mebel *et al.* (1996b)
R1514	$NO + NCO$	$\longrightarrow$	$N_2 + CO_2$	$9.63 \times 10^{-6}\, T^{-1.97}\, e^{(-560/T)}$	Lin *et al.* (1993)
R1516	$NO + NCO$	$\longrightarrow$	$N_2O + CO$	$1.06 \times 10^{-7}\, T^{-1.34}\, e^{(-360/T)}$	Lin *et al.* (1993)
R1518	$NO_2 + H$	$\longrightarrow$	$NO + OH$	$2.2 \times 10^{-10}\, e^{(-182/T)}$	Ko and Fontijn (1991)
R1520	$NO_2 + CH$	$\longrightarrow$	$NO + HCO$	1.67×10^{-10}	Wagal *et al.* (1982)
R1522	$NO_2 + {}^3CH_2$	$\longrightarrow$	$NO + H_2CO$	9.8×10^{-11}	Seidler *et al.* (1989)
R1524	$NO_2 + CH_3$	$\longrightarrow$	$NO + CH_3O$	2.3×10^{-11}	Biggs *et al.* (1993)
R1526	$NO_2 + CH_4$	$\longrightarrow$	$HNO_2 + CH_3$	$1.99 \times 10^{-11}\, e^{(-15097/T)}$	Slack and Grillo (1981)
R1528	$NO_2 + O$	$\longrightarrow$	$NO + O_2$	$6.5 \times 10^{-12}\, e^{(120/T)}$	Tsang and Herron (1991)
R1530	$NO_2 + O(^1D)$	$\longrightarrow$	$NO + O_2$	3.0×10^{-10}	Atkinson *et al.* (1992)
R1532	$NO_2 + CO$	$\longrightarrow$	$NO + CO_2$	$1.5 \times 10^{-10}\, e^{(-17000/T)}$	Tsang and Herron (1991)
R1534	N_2O	$\xrightarrow{M}$	$N_2 + O$	$k_0 = 5.88 \times 10^{-10}\, e^{(-28265/T)}$	Breshears (1995)
				$k_\infty = 1.3 \times 10^{11}\, e^{(-30000/T)}$	Tsang and Herron (1991)
R1536	$N_2O + H$	$\longrightarrow$	$N_2 + OH$	$1.1 \times 10^{-31}\, T^{6.03}\, e^{(-800/T)}$	literature review
R1538	$N_2O + H$	$\longrightarrow$	$N_2H + O$	$9.14 \times 10^{-6}\, T^{-1.06}\, e^{(-23800/T)}$	Bozzelli *et al.* (1994)
R1540	$N_2O + CH$	$\longrightarrow$	$HCN + NO$	6.9×10^{-11}	Becker *et al.* (1989)

TABLE S1 (*cont.*)

	Reaction			Rate constant	Reference
R1542	$N_2O + O$	$\longrightarrow$	$N_2 + O_2$	$1.7 \times 10^{-10}\, e^{(-14100/T)}$	Tsang and Herron (1991)
R1544	$N_2O + O$	$\longrightarrow$	$2\,NO$	$1.1 \times 10^{-10}\, e^{(-13400/T)}$	Tsang and Herron (1991)
R1546	$N_2O + O(^1D)$	$\longrightarrow$	$2\,NO$	7.2×10^{-11}	Atkinson *et al.* (1992)
R1548	$N_2O + OH$	$\longrightarrow$	$N_2 + HO_2$	$2.15 \times 10^{-26}\, T^{4.72}\, e^{(-18400/T)}$	Mebel *et al.* (1996b)
R1550	$N_2O + CO$	$\longrightarrow$	$N_2 + CO_2$	$5.3 \times 10^{-13}\, e^{(-10230/T)}$	Tsang and Herron (1991)
R1552	$HNO + H$	$\longrightarrow$	$NO + H_2$	$3.0 \times 10^{-11}\, e^{(-500/T)}$	Tsang and Herron (1991)
R1554	$HNO + H_2$	$\longrightarrow$	$NH + H_2O$	$1.66 \times 10^{-10}\, e^{(-8055/T)}$	Rohrig and Wagner (1994b)
R1556	$HNO + CH_3$	$\longrightarrow$	$NO + CH_4$	$1.36 \times 10^{-18}\, T^{1.87}\, e^{(-480/T)}$	Dean and Bozzelli (2000)
R1558	$HNO + O$	$\longrightarrow$	$NO + OH$	6.0×10^{-11}	Tsang and Herron (1991)
R1560	$HNO + O_2$	$\longrightarrow$	$NO + HO_2$	$3.65 \times 10^{-14}\, e^{(-4600/T)}$	Bryukov *et al.* (1993)
R1562	$HNO + OH$	$\longrightarrow$	$NO + H_2O$	$8.0 \times 10^{-11}\, e^{(-500/T)}$	Tsang and Herron (1991)
R1564	$2\,HNO$	$\longrightarrow$	$N_2O + H_2O$	$1.4 \times 10^{-15}\, e^{(-1550/T)}$	He and Lin (1992)
R1566	$HNO_2 + H$	$\longrightarrow$	$NO + H_2O$	$1.35 \times 10^{-17}\, T^{1.89}\, e^{(-1935/T)}$	Dean and Bozzelli (2000)
R1568	$HNO_2 + H$	$\longrightarrow$	$NO_2 + H_2$	$3.3 \times 10^{-16}\, T^{1.55}\, e^{(-3330/T)}$	Dean and Bozzelli (2000)
R1570	$HNO_2 + H$	$\longrightarrow$	$HNO + OH$	$9.3 \times 10^{-14}\, T^{0.86}\, e^{(-2500/T)}$	Dean and Bozzelli (2000)
R1572	$HNO_2 + OH$	$\longrightarrow$	$NO_2 + H_2O$	$2.09 \times 10^{-14}\, T^{1}\, e^{(-68/T)}$	Tsang and Herron (1991)
R1574	NCO	$\xrightarrow{M}$	$N + CO$	$k_0 = 3.65 \times 10^{-10}\, e^{(-27200/T)}$	Mertens and Hanson (1996)
R1576	$NCO + H$	$\longrightarrow$	$NH + CO$	8.9×10^{-11}	Tsang (1992)
R1578	$NCO + H_2$	$\longrightarrow$	$HNCO + H$	$1.26 \times 10^{-21}\, T^{3}\, e^{(-2000/T)}$	Miller and Melius (1992b)
R1580	$NCO + CH_4$	$\longrightarrow$	$HNCO + CH_3$	$1.63 \times 10^{-11}\, e^{(-4090/T)}$	Schuck *et al.* (1994)
R1582	$NCO + C_2H_6$	$\longrightarrow$	$HNCO + C_2H_5$	$2.4 \times 10^{-33}\, T^{6.89}\, e^{(-1467/T)}$	Schuck *et al.* (1994)
R1584	$NCO + O$	$\longrightarrow$	$N + CO_2$	$1.3 \times 10^{-11}\, e^{(-1300/T)}$	Dean and Bozzelli (2000)
R1586	$NCO + O$	$\longrightarrow$	$NO + CO$	7.5×10^{-11}	Tsang (1992)
R1588	$HNCO$	$\xrightarrow{M}$	$NH + CO$	$k_0 = 3.6 \times 10^{4}\, T^{-3.1}\, e^{(-51300/T)}$	Tsang (1992)
				$k_\infty = 6.0 \times 10^{13}\, e^{(-50228/T)}$	Tsang (1992)
R1590	$HNCO + H$	$\longrightarrow$	$NH_2 + CO$	$5.96 \times 10^{-20}\, T^{2.49}\, e^{(-1180/T)}$	Nguyen *et al.* (1996)
R1592	$HNCO + O$	$\longrightarrow$	$NCO + OH$	$3.65 \times 10^{-18}\, T^{2.11}\, e^{(-5750/T)}$	He *et al.* (1992)
R1594	$HNCO + O$	$\longrightarrow$	$NH + CO_2$	$1.63 \times 10^{-16}\, T^{1.41}\, e^{(-4290/T)}$	He *et al.* (1992)
R1596	$HNCO + OH$	$\longrightarrow$	$NCO + H_2O$	$1.06 \times 10^{-18}\, T^{2}\, e^{(-1290/T)}$	Tsang (1992)
R1598	$HNCO + OH$	$\longrightarrow$	$NH_2 + CO_2$	$2.99 \times 10^{-18}\, T^{1.5}\, e^{(-1809/T)}$	Woodbridge *et al.* (1996)
R1600	$HNCO + HCO$	$\longrightarrow$	$NCO + H_2CO$	$5.0 \times 10^{-12}\, e^{(-13100/T)}$	Tsang (1992)
R1602	$CH_3NO + H$	$\longrightarrow$	$HNO + CH_3$	$3.0 \times 10^{-11}\, e^{(-1400/T)}$	Dean and Bozzelli (2000)
R1604	$CH_3NO + O$	$\longrightarrow$	$NO_2 + CH_3$	$2.82 \times 10^{-18}\, T^{2.08}$	Dean and Bozzelli (2000)
R1606	$CH_3NO + OH$	$\longrightarrow$	$HNO_2 + CH_3$	$4.15 \times 10^{-12}\, e^{(-500/T)}$	Dean and Bozzelli (2000)

Only the rate coefficients for the forward reactions are presented in the table. Reverse reactions immediately follow the forward reactions in the full list such that forward and reverse reactions are paired together, with the reverse reactions having odd numbers. Rate coefficients for the reverse reactions are internally reversed using the thermodynamic principle of microscopic reversibility; see the full model results for these rate coefficients. The phrase "Test case" refers to sensitivity tests that are not discussed in the paper. The term "M" in a reaction represents any third body, such as H_2. All temperatures are in kelvins. Rate coefficients for binary reactions (e.g., A + B → products), high-pressure limiting rate coefficients (k_∞) for three-body reactions (e.g., A + B + M → AB + M), and low-pressure limiting rate coefficients (k_0) for decomposition reactions (e.g., AB + M → A + B + M) are in units of $cm^3\ s^{-1}$. Low-pressure limiting rate coefficients for three-body reactions (e.g., k_0 for reactions of the type A + B + M → AB + M) are in units of $cm^6\ s^{-1}$. Unimolecular decomposition reactions and k_∞ for AB + M → A + B + M reactions have units of s^{-1}.

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